MIL-M-28787/154A(EC)
22 January 1979
SUPERSEDING
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MILITARY SPECIFICATION

MODULES, STANDARD ELECTRONIC

LOGIC CONVERTER, ANALOG TO DIGITAL, KEY CODE DBA

This specification is approved for use by Naval Electronic Systems Command, Department of the Navy, and is available for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 Scope. This document establishes the requirements and test procedures for the procurement of an electronic analog to digital logic converter module.

1.2 Module description. The following electrical parameters reflect the 0° to 60°C end-of-life limits from table III.

Name	Logic converter, analog to digital
Family	Analog to digital
Key code	DBA
Size	1A
Weight	35 grams maximum
Failure rate	3.02 failures/10^6 hours maximum
Environment	Class I as specified in MIL-STD-1389
Supply voltage	5.0 Vdc ±10 percent
Maximum power dissipation	2.12 W (outputs not loaded)
Input voltage	
Low level	-0.5 V minimum, 0.7 V maximum
High level	2.0 V minimum, V_{CC} maximum
Input current, low level V_{IL} = 0.0 V	
Comparator	1.73 mA maximum
Reset	5.19 mA maximum
Clock	3.46 mA maximum
Conversion control	3.46 mA maximum
Convert	3.46 mA maximum
Input current, low level V_{IL} = 0.4 V	
Comparator	1.6 mA maximum
Reset	4.8 mA maximum
Clock	3.2 mA maximum
Conversion control	3.2 mA maximum
Convert	3.2 mA maximum

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Input current, high level, V_{IH} = 2.4 V	
Comparator	40 µA maximum
Reset	120 µA maximum
Clock	80 µA maximum
Conversion control	80 µA maximum
Convert	80 µA maximum
Input current, high level, V_{IH} = 5.5 V	
Comparator	1.0 mA maximum
Reset	1.0 mA maximum
Clock	2.0 mA maximum
Conversion control	2.0 mA maximum
Convert	2.0 mA maximum
Output voltage, low level	
Output (I_{OL} = 8.0 mA)	0.4 V maximum
$\overline{\text{Output}}$ (I_{OL} = 9.6 mA)	0.4 V maximum
Serial (I_{OL} = 16 mA)	0.4 V maximum
Conversion in progress (I_{OL} = 8.0 mA)	0.4 V maximum
Output voltage, high level	
Output (I_{OH} = -400 µA)	2.4 V minimum
$\overline{\text{Output}}$ (I_{OH} = -480 µA)	2.4 V minimum
Serial (I_{OH} = -400 µA)	2.4 V minimum
Conversion in progress (I_{OH} = -100 µA)	2.4 V minimum
* Output drive current maximum capability	
Low level	
Output (V_{OL} = 0.4 V)	8.0 mA maximum
$\overline{\text{Output}}$ (V_{OL} = 0.4 V)	9.6 mA maximum
Serial (V_{OL} = 0.4 V)	16 mA maximum
Conversion in progress (V_{OL} = 0.4 V)	8.0 mA maximum
High level	
Output (V_{OH} = 2.4 V)	-400 µA maximum
$\overline{\text{Output}}$ (V_{OH} = 2.4 V)	-480 µA maximum
Serial (V_{OH} = 2.4 V)	-400 µA maximum
Conversion in progress (V_{OH} = 2.4 V)	-100 µA maximum
Comparator input setup time	0 ns
Comparator input hold time	100 ns
Propagation delay time	
From clock to BIT	
t_{PLH}	100 ns minimum, 650 ns maximum
t_{PHL}	10 ns minimum, 650 ns maximum

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	Propagation delay time -Continued	
	From clock to $\overline{BIT}$	
	t_{PLH}	10 ns minimum, 680 ns maximum
	t_{PHL}	100 ns minimum, 680 ns maximum
	From clock to SER	
	t_{PLH}	10 ns minimum, 110 ns maximum
	t_{PHL}	10 ns minimum, 125 ns maximum
*	From reset to BIT 1	
	t_{PHL}	30 ns minimum, 300 ns maximum
*	From clock to CIP	
	t_{PLH}	30 ns minimum, 225 ns maximum
	From reset to CIP	
	t_{PHL}	5 ns minimum, 90 ns maximum
	From reset to SER	
	t_{PHL}	10 ns minimum, 130 ns maximum

1.3 Module function. The 12-BIT analog-to-digital logic converter module, designated as key code DBA, is a Standard Electronic Modules Program (SEMP) electronic module with 0.100 inch (2.54 mm) grid center contacts. This module contains successive approximation analog-to-digital converter logic. It is designed to be used with the 12-BIT digital-to-analog converter module (DBC). When used with the DBC module the A/D converter system has a maximum clock frequency of 280 kHz. Three analog-to-digital conversion wiring schemes are illustrated on figure 1. This module is capable of dc, pulse, and continuous trigger modes. These three modes will be illustrated assuming 12-BIT conversions and the reset (pin 17) connected to $\overline{BIT\ 14}$ (pin 21). In all modes the parallel data outputs are valid after 13 clock periods and the parallel data is held for 1 clock period to a low level in the dc convert mode. Three conversion timing diagrams are illustrated on figures 2, 3, and 4.

1.3.1 Conversion control (pin 35) and convert (pin 37) inputs. If the conversion control input (pin 35) and the convert input (pin 37) are at a high level, the A/D operation is in continuous convert mode. The data is only valid during each 13th clock period. On the 14th clock pulse the conversion in progress is set to a low level. On the 15th clock pulse all the outputs are set to a low level with the reset (pin 17) connected to $\overline{BIT\ 14}$ (pin 21). On the 16th clock pulse a new conversion starts. If the conversion control input (pin 35) is at a high level and the convert input (pin 37) is pulsed (pulse width is less than 13 clock pulses), the A/D operation is in the ac convert mode. The data is only valid during the 13th clock period. On the 14th clock pulse with the reset (pin 17) connected to $\overline{BIT\ 14}$ (pin 21), the conversion in progress (pin 15) goes to a low level. On the 15th clock pulse the converter is completely reset. If the conversion control input (pin 35) is at a low level and the convert input (pin 37) goes from a low level to a high level after the converter is reset, the A/D operation is in the dc convert mode. The data is valid after the rising edge of the 13th clock pulse. When the reset (pin 17) is connected to $\overline{BIT\ 14}$ (pin 21) the conversion in progress goes to low level after the 14th clock pulse but the data remains valid until the convert input goes from a high level to low level and then all outputs are reset.

1.3.2 Comparator (pin 19) input. When the comparator input (pin 19) is at a high level, the output will not change state in the next clock pulse. Therefore, on the 13th clock pulse all outputs will be at a high level. When the comparator input (pin 19) is at a low level, the output will change state in the next clock pulse. Therefore, on the 13th clock pulse all outputs will be at a low level.

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1.3.3 Reset (pin 17) input. Reset functions will occur within two clock pulses for the ac and continuous convert modes.

1.3.4 Pin programming for voltage scales. The pin programming for 0 to 10 V, 0 to 5 V, -10 to +10 V, and -5 to +5 V operation is done on the companion module. Instructions for that pin programming can be found in the companion module specification and in table I.

TABLE I. Pin programming for voltage scales.

Voltage scale	12-BIT conversion DBC module pin to pin
0 to 10 V	1C18 to 1A17
0 to 5 V	1C18 to 1A17, 1C19 to 1C17
-10 to +10 V	1A7 to 1C19, 1C17 to 1A17
-5 to +5 V	1A7 to 1C19, 1C18 to 1A17

NOTE. For the arrangement of pins on the DBC module see figure 1 or the companion module specification.

1.3.5 Input to output relationship. When the module is used with its companion module for analog-to-digital conversion, the binary number on the digital outputs will correspond to the analog voltage put into the converter. As the analog input voltage increases by a least significant BIT increment, the binary number on the digital outputs will increase by one. Least significant bit values are determined by the companion module. The values for the least significant BIT are listed in table II.

TABLE II. Least significant BIT values.

Voltage range	Least significant BIT	
	8-BIT conversion	12-BIT conversion
0 to 10 V	0.0391 V	0.02441 V
0 to 5 V	0.0195 V	0.00122 V
-10 to +10 V	0.0781 V	0.00488 V
-5 to +5 V	0.0391 V	0.02441 V

2. APPLICABLE DOCUMENTS

2.1 Issues of documents. The following documents of the issue in effect on date of invitation for bids or request for proposal, form a part of this specification to the extent specified herein.

SPECIFICATION

MILITARY

MIL-M-28787 - Modules, Standard Electronic, General Specification for.

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STANDARDS

MILITARY

MIL-STD-1389 - Design Requirements for Standard Electronic Modules.

MIL-STD-1665 - Test Equipment for the Standard Electronic Modules Program.

(All military standards and military specifications shall be ordered from the Naval Publications and Forms Center, 5801 Tabor Avenue, Philadelphia, PA 19120.)

3. REQUIREMENTS

3.1 General requirements. The module shall comply with the requirements of MIL-M-28787 except as modified herein.

3.2 Correlation requirements. Before any production modules are tested, all testers and test equipment shall be correlated in accordance with MIL-M-28787.

3.3 Failure rate requirements. All modules procured to this specification shall have a predicted failure rate no greater than 3.02 failures per million operating hours when predicted in accordance with MIL-STD-1389.

3.4 Performance requirements. The module shall conform to the electrical requirements specified in table III and shall operate within the environment of design class I as specified in MIL-STD-1389.

3.5 Module configuration requirements. The module configuration requirements shall be in accordance with MIL-STD-1389 and as specified herein.

3.5.1 Key code. The module key code shall be DBA.

3.5.2 Size. The module size shall be 1A.

3.5.3 Weight. The module shall have a maximum weight of 35 grams.

3.5.4 Contact pins. The module shall have 40 contact pins positioned A1 through A40.

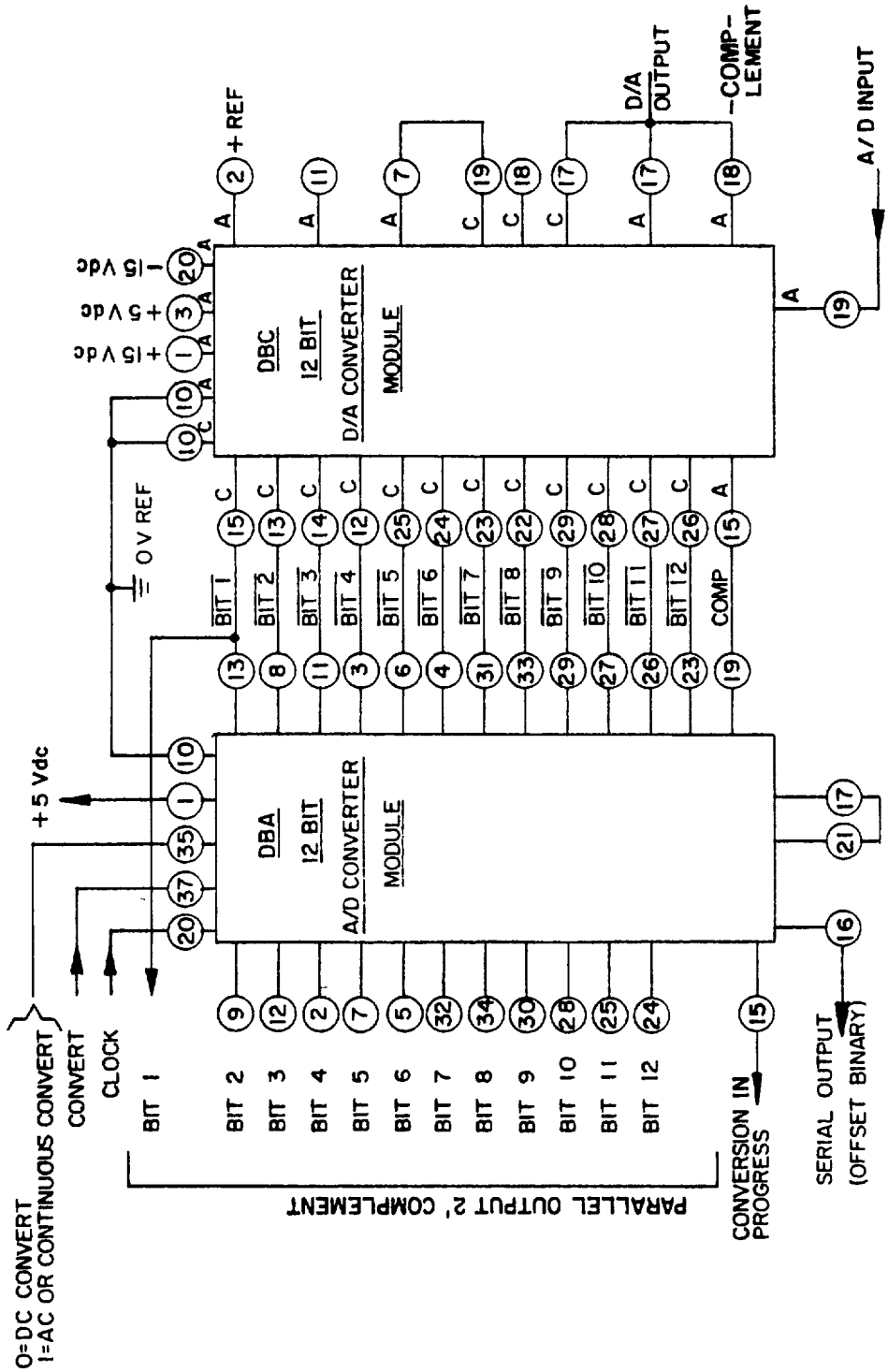
3.5.5 Fin position. The module shall have an insertion and extraction and heat dissipation fin on increment A.

3.5.6 Depth. The module shall have no additional depth requirements.

3.5.7 Holddown holes. The module shall have no additional requirements for location or quantity of holddown holes.

3.5.8 Marking. Marking shall be as specified in MIL-STD-1389, and the module shall be marked with the military part number M28787/154, except no marking shall interfere with heat dissipation from the fin (for example, affixing a label which may be a heat barrier).

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NOTE: Pin programming is available on the DBC module for unipolar operation.

FIGURE 1. 12 BIT bipolar A/D converter.

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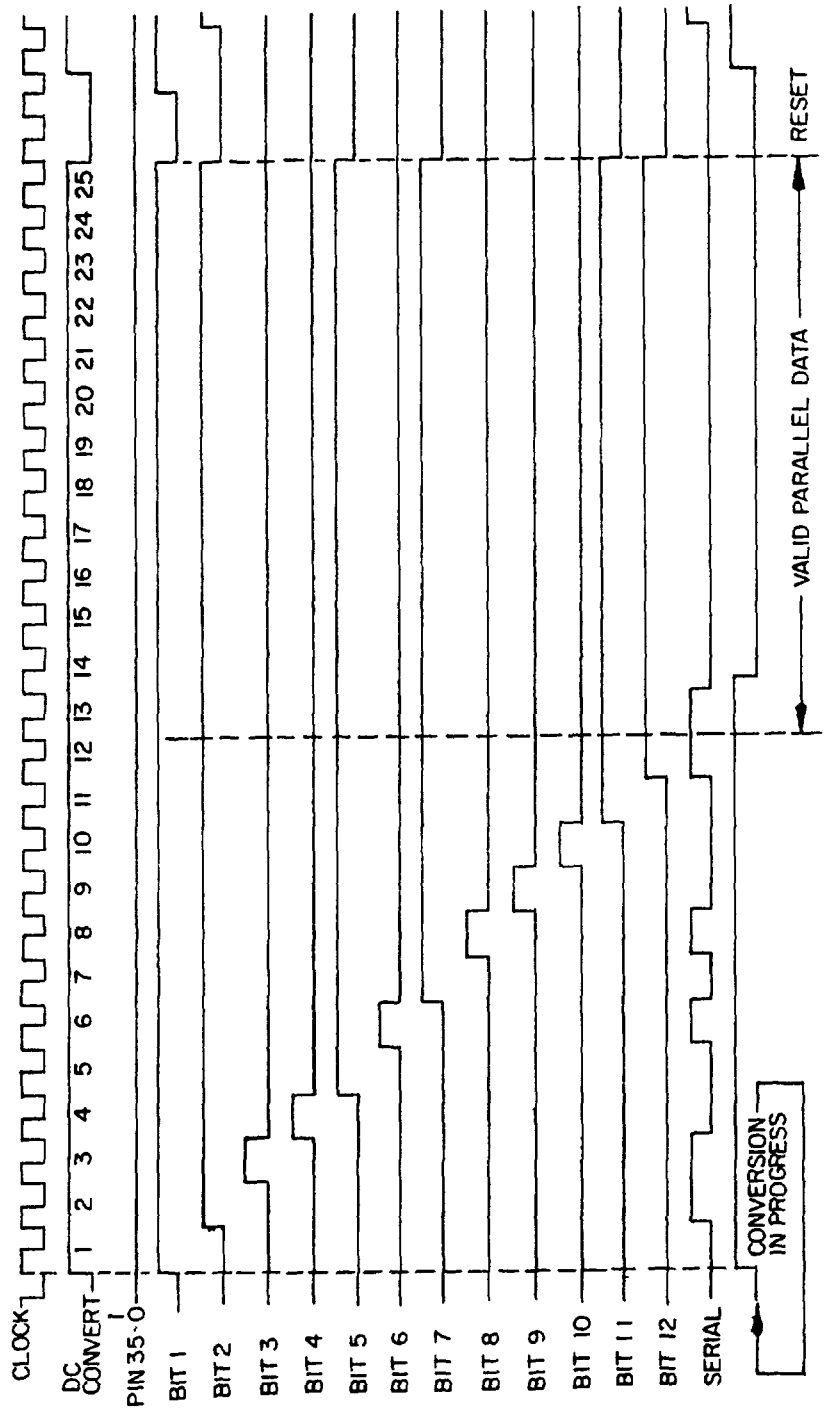


FIGURE 2. Timing diagram 1, dc convert mode.

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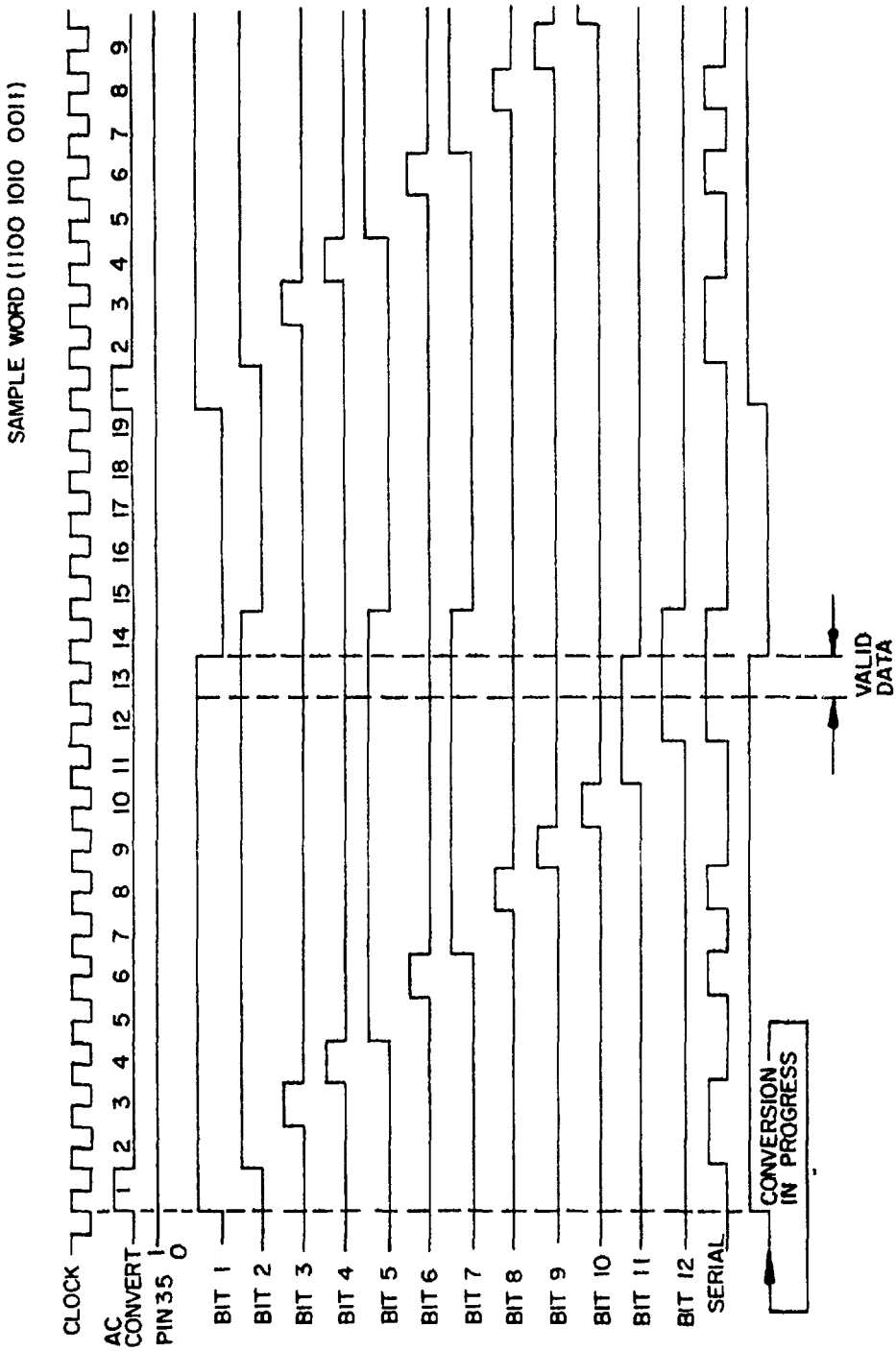


FIGURE 3. Timing diagram 2, ac convert mode.

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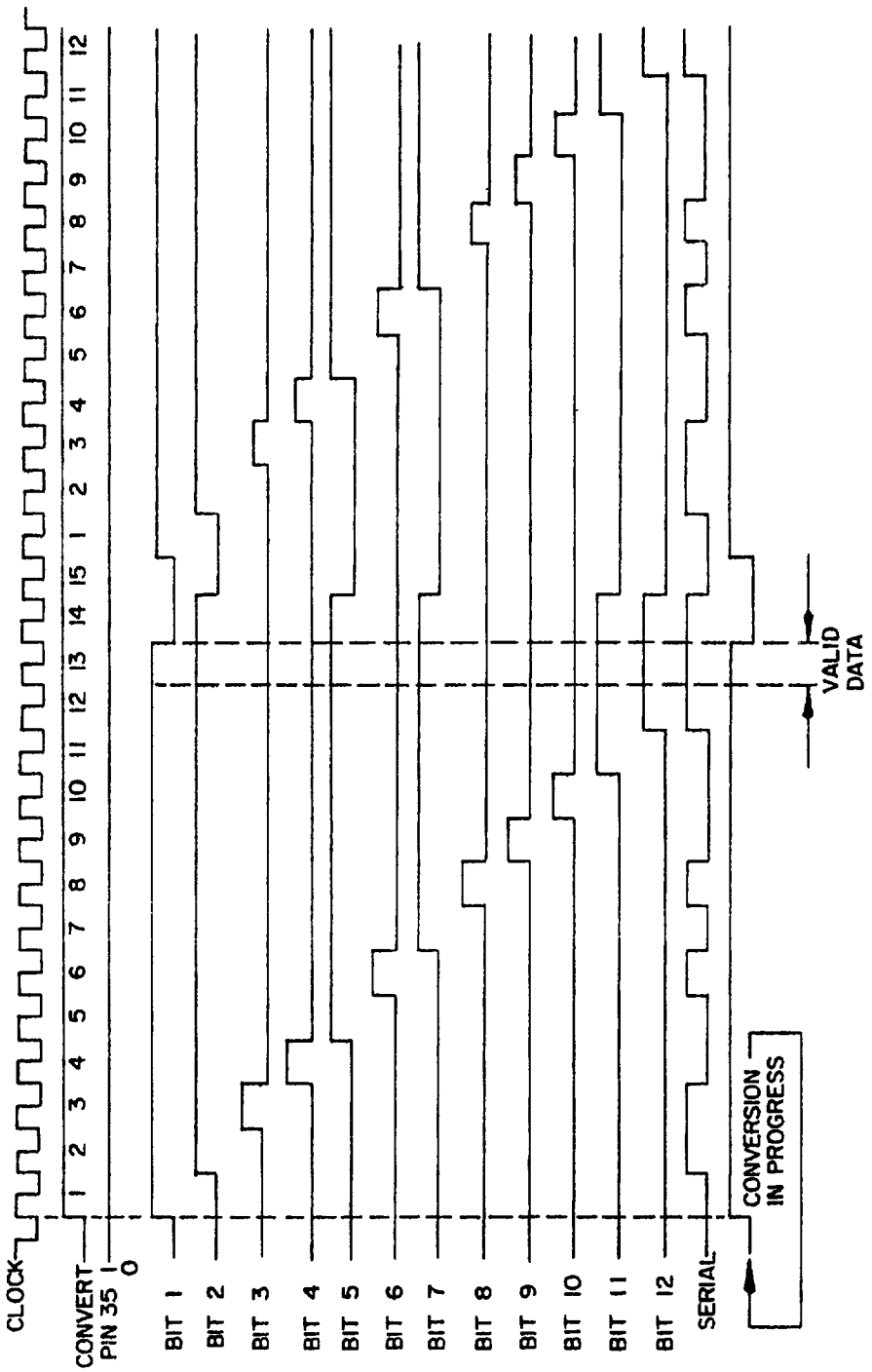


FIGURE 4. Timing diagram 3, continuous convert mode.

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3.6 Circuit pin configuration. The function diagram and pin assignments shall conform to figure 5.

3.7 Acceptance test. All production modules shall be tested in accordance with notes 1 and 2 of table III.

3.8 End-of-life limits. Modules which have been plugged into an operating system, or exceed two years from date of manufacture and are to be tested, shall be capable of conforming to the 25°C end-of-life limits of table III.

3.9 Power supply voltage. The module shall be capable of conforming to the requirements of table III for a power supply voltage (V_{CC}) of 5.0 ±0.5 Vdc, except that the limits for output short circuit current apply only for a power supply voltage of 5.5 Vdc ±1 percent.

4. QUALITY ASSURANCE PROVISIONS

4.1 Quality assurance. The quality assurance provisions of MIL-M-28787 shall apply.

4.2 Acceptance tests. Acceptance tests of all modules shall be in accordance with procedures and equipment specified in MIL-M-28787.

4.3 Test equipment. The following test equipment is necessary to conduct acceptance tests for this module. Test equipment shall be as specified in MIL-STD-1665 and that listed below:

Equipment	MIL-STD-1665 designator	No. required (maximum per test figure)
DC power supply	Item P01	4
Pulse generator	Item J01	2
DC voltmeter	Item V03	1
Oscilloscope	Item D05	1
Oscilloscope probe	Item Y01	2
DC ammeter	Item X01	1
Megohm bridge	Item R06	1
Capacitance bridge	Item C01	1
Capacitance meter	Item C05	1
Resistors		
634Ω ±1%, 1/8 W		13
536Ω ±1%, 1/8 W		14
316Ω ±1%, 1/8 W		1
1 kΩ ±1%, 1/8 W		1
51.1Ω ±5%, 1/2 W		3
6.04 kΩ ±1%, 1/8 W		1
23.7 kΩ ±1%, 1/8 W		1
4.99 kΩ ±1%, 1/8 W		1
Value as required for figure 14, note 4b, 1/8 W		1
Value as required for figure 14, note 5b, 1/8 W		1
Value as required for figure 14, note 6b, 1/8 W		1
Value as required for figure 14, note 7b, 1/8 W		1
Capacitors		
Value as required for figure 14, note 4c		1
Value as required for figure 14, note 5c		1
Value as required for figure 14, note 6c		1
Value as required for figure 14, note 7c		1

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Diode	No. required (maximum per test figure)
1N4150, or equivalent	4
Switches	
SPST	1
SPDT	1
Other	
Logic state indicators as required for figure 14, note 8	14

All resistors shall be metal film and all capacitors shall be ceramic unless otherwise specified.

4.4 Test procedures. The module shall be tested as specified in table III.

4.4.1 Alternate test methods. Alternate test methods may be employed if approved by the SEMP Quality Assurance Activity (SEMP-QAA) and they conform to the requirements of MIL-M-28787.

4.4.2 Order of testing. Test conditions A through W shall be performed in any order with the following exceptions. If conditions A, B, or C are part of a sequence of tests, the order of testing shall be as follows: A, B, C where applicable, followed by the remaining tests in the sequence.

4.5 Qualification testing. Initial qualification tests and periodic-check tests shall be performed in accordance with the test plan specified in MIL-M-28787 with the following modifications or additions.

* 4.5.1 Operating temperature tests. The operating temperature tests shall be conducted in accordance with MIL-M-28787. Before starting these tests, test conditions A through W in table III shall be performed in the order specified in 4.4.2 using the initial limits at 25° ±5°C ambient. When electrical testing is called for at the end of each temperature stabilization period, test conditions D through U shall be performed in the order specified in 4.4.2 using the initial limits for applicable operating temperature. At all other times during the operating temperature tests, the module shall be operated in the test circuit shown on figure 6.

4.5.1.1 Component temperature test. The component temperature test shall be conducted in accordance with MIL-M-28787. The module shall be operated as shown on figure 7.

4.5.2 Vibration tests. The vibration tests shall be conducted in accordance with MIL-M-28787. During the vibration tests, the module shall be operated in the test circuit shown on figure 8. Following the vibration tests, the electrical tests of table III which are marked with an (X) in the test type column shall be performed using the initial limits at 25° ±5°C ambient.

* 4.5.3 Life test. The life test shall be conducted in accordance with MIL-M-28787. The module shall be operated as shown on figure 6. Following the life test and while still at the maximum class temperature, electrical tests D through U shall be performed in the order specified in 4.4.2 using the operating temperature end-of-life requirements of table III. Following the return of the module to 25°C, electrical tests D through W shall be performed in the order specified in 4.4.2 using the 25° ±5°C ambient end-of-life requirements of table III.

5. PACKAGING

5.1 Packaging requirements. The requirements for packaging shall be in accordance with MIL-M-28787.

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6. NOTES

6.1 Definitions. Definitions are as specified in MIL-STD-1331.

6.2 End-of-life. When any individual module circuit parameter fails to conform to the end-of-life requirements of table III, the module is considered to have completed its useful life.

6.3 Changes from previous issue. The margins of this specification are marked with an asterisk to indicate where changes (additions, modifications, corrections, deletions) from the previous issue were made. This was done as a convenience only and the Government assumes no liability whatsoever for any inaccuracies in these notations. Bidders and contractors are cautioned to evaluate the requirements of this document based on the entire content irrespective of the marginal notations and relationship to the last previous issue.

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TABLE III. Module electrical requirements.

Test condition	Parameter	Ref test table	Test type (see notes)	Acceptance requirements 25° ±5°C Ambient temperature				Operating temperature 0° to 60°C				Unit
				Initial		End-of-life		Initial		End-of-life		
				Min	Max	Min	Max	Min	Max	Min	Max	
A1-A28	V_{out} (low-loaded)	IV	S	0.1	0.8	0.1	0.8	0.1	0.8	0.1	0.8	V
B1, B2	Input breakdown current (R, CM)	V	S	0.0	1.0	0.0	1.0	0.0	1.0	0.0	1.0	mA
B3-B5	Input breakdown current (CP, CV, CT)	V	S	0.0	2.0	0.0	2.0	0.0	2.0	0.0	2.0	mA
C1, C2	Isolation	VI	S	10.0	∞	1.0	∞	10.0	∞	1.0	∞	MΩ
D1	Input current, low level (R)	VII	X	0.01	5.19	0.01	5.19	0.01	5.19	0.01	5.19	mA
D2-D4	Input current, low level (CP, CV, CT)	VII	X	0.1	3.46	0.1	3.46	0.1	3.46	0.1	3.46	mA
D5	Input current, low level (CM)	VII	X	0.1	1.73	0.1	1.73	0.1	1.73	0.1	1.73	mA
E1	Input current, high level (R)	VII	X	0	120	0	120	0	120	0	120	µA
E2-E4	Input current, high level (CP, CV, CT)	VII	X	0	80	0	80	0	80	0	80	µA
E5	Input current, high level (CM)	VII	X	0	40	0	40	0	40	0	40	µA
F1, F2	Output short circuit current (CIP, SER)	VII	X	1	100	1	100	1	100	1	100	mA
F3-F16	Output short circuit current ($\overline{BIT1}$-$\overline{BIT14}$)	VII	X	10	100	10	100	10	100	10	100	mA
F17-F28	Output short circuit current (BIT1-BIT12)	VII	X	20	100	20	100	20	100	20	100	mA
G1-G14	Output voltage, high level	VIII	X	2.4	4.5	2.4	4.5	2.4	4.5	2.4	4.5	V
H1-H14	$\overline{\text{Output}}$ voltage, low level	VIII	X	0.0	0.4	0.0	0.4	0.0	0.4	0.0	0.4	V
J1-J14	Output voltage, low level	VIII	X	0.0	0.4	0.0	0.4	0.0	0.4	0.0	0.4	V
K1-K14	$\overline{\text{Output}}$ voltage, high level	VIII	X	2.4	4.5	2.4	4.5	2.4	4.5	2.4	4.5	V
L	Power supply current	IX	X	100	385	100	385	100	385	100	385	mA

See notes at end of table.

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TABLE III. Module electrical requirements - Continued.

	Test condition	Parameter	Ref test table	Test type (see notes)	Acceptance requirements 25° ±5°C Ambient temperature				Operating temperature 0° to 60°C				Unit
					Initial		End-of-life		Initial		End-of-life		
					Min	Max	Min	Max	Min	Max	Min	Max	
	M1-M33	Functional A/D conversion, DC convert	X	X	← All tests must pass →								
	N1-N31	Functional A/D conversion, AC convert	X	X	← All tests must pass →								
	P1-P31	Functional A/D conversion, continuous convert	X	X	← All tests must pass →								
	Q1-Q26	Output transition time, t_{TLH}	XI	S	5	50	5	50	5	55	5	55	ns
	R1-R26	Output transition time, t_{THL}	XI	S	2	20	2	20	2	25	2	25	ns
	S1-S23 (odd)	Propagation delay time (t_{PLH}), BIT, and setup and hold time check	XII	S	100	600	100	600	100	650	100	650	ns
	T1-T23 (odd)	Propagation delay time (t_{PHL}), BIT, and setup and hold time check	XII	S	10	600	10	600	10	650	10	650	ns
	S2-S24 (even)	Propagation delay time (t_{PLH}), $\overline{BIT}$, and setup and hold time check	XII	S	10	620	10	620	10	680	10	680	ns
	T2-T24 (even)	Propagation delay time (t_{PHL}), $\overline{BIT}$, and setup and hold time check	XII	S	100	620	100	620	100	680	100	680	ns
	S25	Propagation delay time (t_{PLH}), SFR	XII	S	10	100	10	100	10	110	10	110	ns
	T25	Propagation delay time (t_{PHL}), SER	XII	S	10	115	10	115	10	125	10	125	ns
*	T26	Propagation delay time (t_{PHL}), R to BIT1	XII	S	30	250	30	250	30	300	30	300	ns
*	S26	Propagation delay time (t_{PLH}), CP to CIP	XII	S	30	200	30	200	30	225	30	225	ns
	T27	Propagation delay time (t_{PHL}), R to CIP	XII	S	5	80	5	80	5	90	5	90	ns
	T28	Propagation delay time (t_{PHL}), R to SER	XII	S	10	120	10	120	10	130	10	130	ns

See notes at end of table.

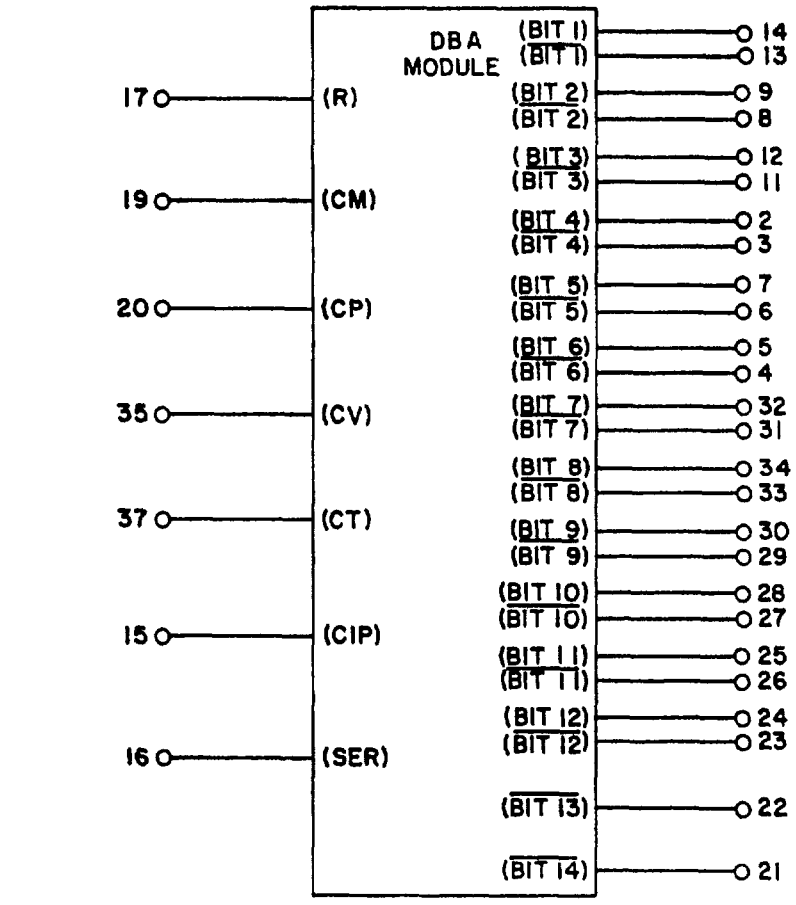
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TABLE III. Module electrical requirements - Continued.

Test condition	Parameter	Ref test table	Test type (see notes)	Acceptance requirements 25° ±5°C Ambient temperature				Operating temperature 0° to 60°C				Unit
				Initial		End-of-life		Initial		End-of-life		
				Min	Max	Min	Max	Min	Max	Min	Max	
U	Power supply filter capacitance	XIII	S	1.6	3.0	1.6	3.0	1.6	3.0	1.6	3.0	µF
V	Power supply transient amplitude	XIV	Q	0	400	0	400	0	400	0	400	mV
W1-W4	Input capacitance	XV	Q	2	25	2	25	2	25	2	25	pF
W5	Input capacitance (CM)	XV	Q	2	15	2	15	2	15	2	15	pF

NOTES:
1. (X) parameters shall be 100 percent tested during module acceptance tests.
2. (S) parameters shall be sample acceptance tested (1 percent or two modules from each inspection lot, whichever is greater) at 25°C.
3. (Q) parameters shall be tested during qualification testing only.

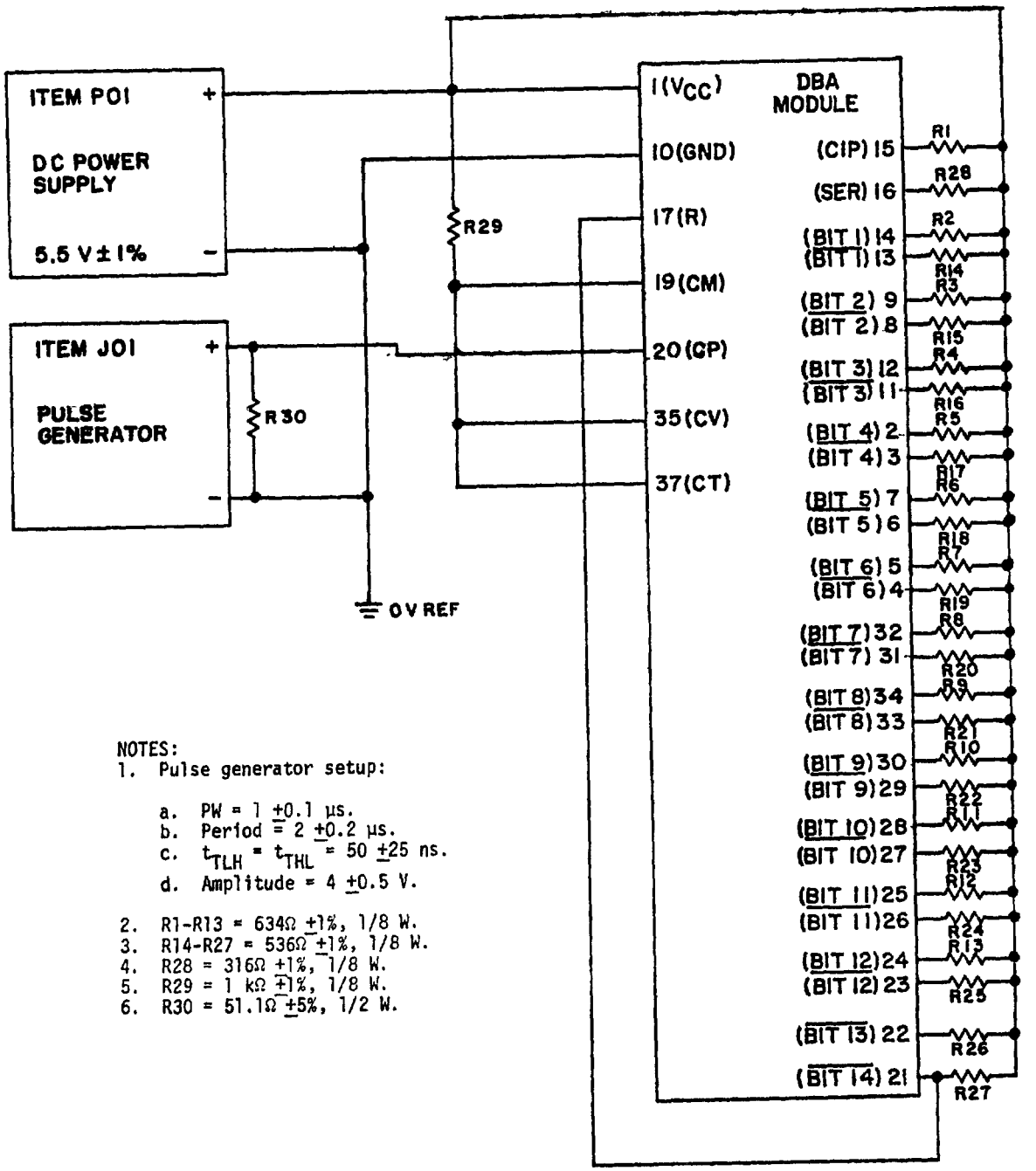
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NOTES:
1. R = Reset input.
2. CM = Comparator input.
3. CP = Clock pulse input.
4. CV = Conversion control input.
5. CT = Convert input.
6. CIP = Conversion in progress output.
7. SER = Serial output.
8. BIT = Digital output.
9. $\overline{\text{BIT}}$ = Inverted digital output.

FIGURE 5. DBA module, 12 BIT analog-to-digital converter.

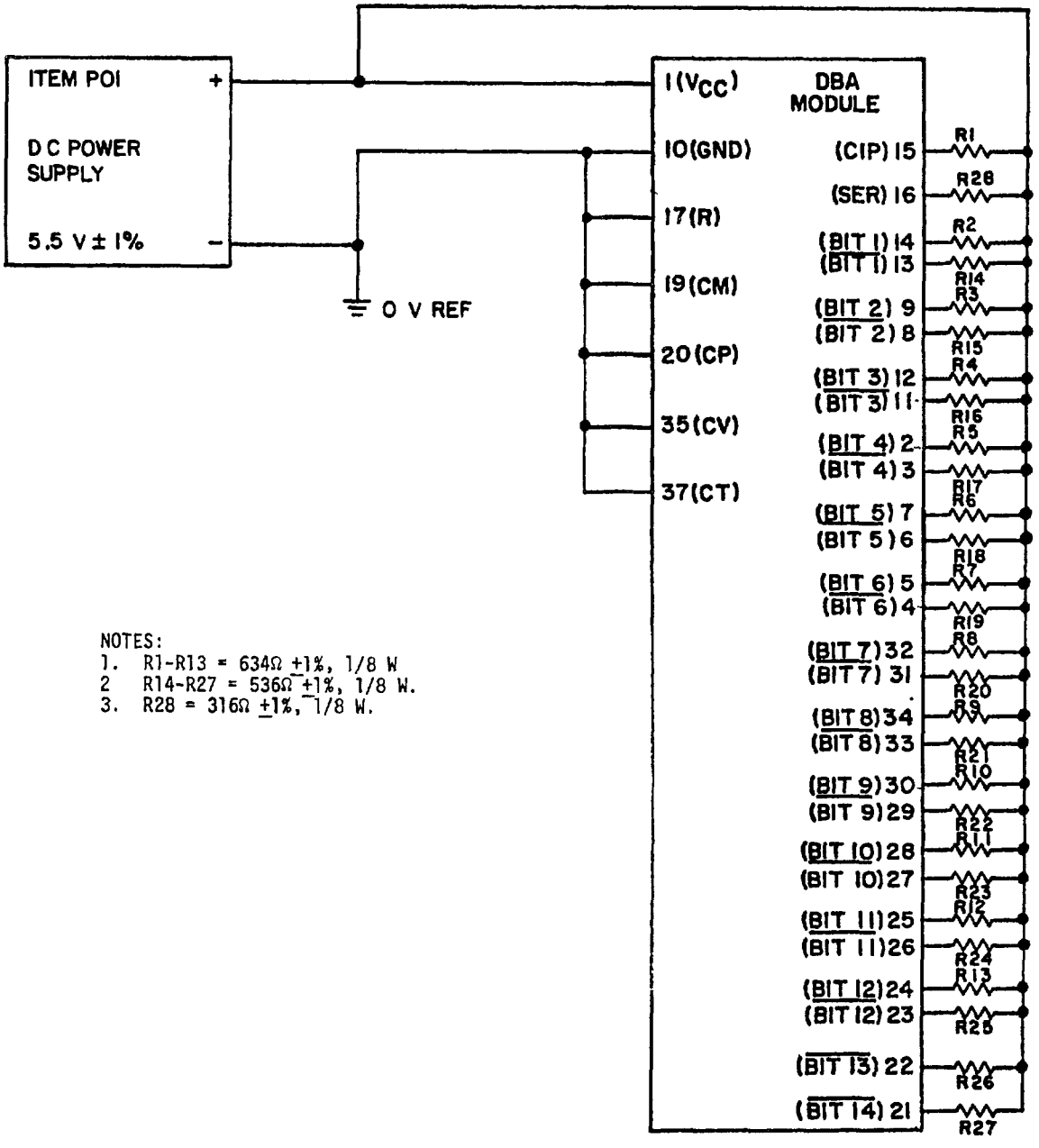
Source: https://assist.dla.mil -- Downloaded: 2026-10-20T16:19Z
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NOTES:
1. Pulse generator setup:
   a. PW = 1 ±0.1 µs.
   b. Period = 2 ±0.2 µs.
   c. $t_{TLH} = t_{THL}$ = 50 ±25 ns.
   d. Amplitude = 4 ±0.5 V.
2. R1-R13 = 634Ω ±1%, 1/8 W.
3. R14-R27 = 536Ω ±1%, 1/8 W.
4. R28 = 316Ω ±1%, 1/8 W.
5. R29 = 1 kΩ ±1%, 1/8 W.
6. R30 = 51.1Ω ±5%, 1/2 W.

FIGURE 6. Operating temperature and life test circuit.

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NOTES:
1. R1-R13 = 634Ω ±1%, 1/8 W
2 R14-R27 = 536Ω ±1%, 1/8 W.
3. R28 = 316Ω ±1%, 1/8 W.

FIGURE 7. Component temperature test circuit.

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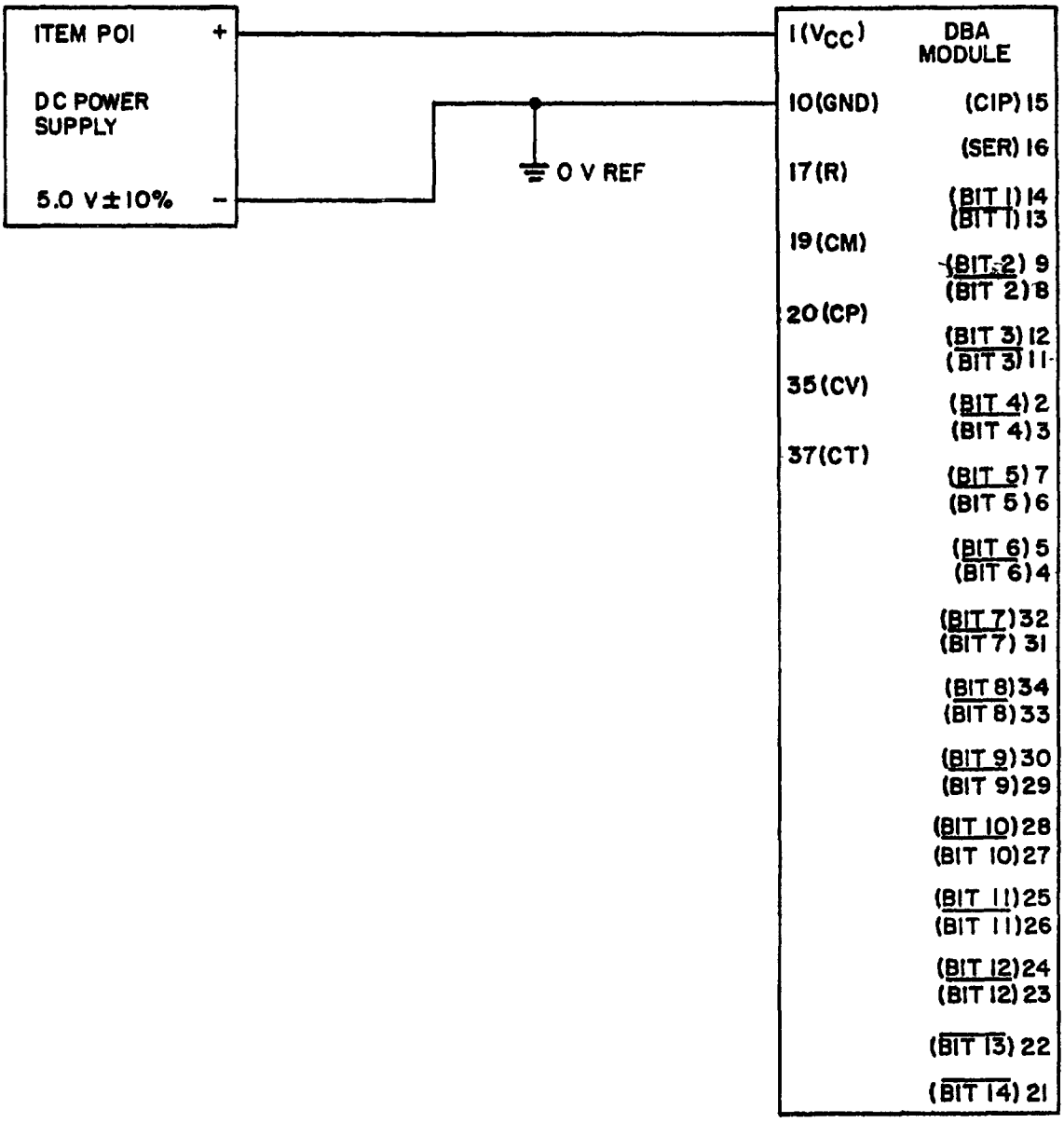


FIGURE 8. Vibration test circuit.

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TABLE IV. V_{out} (low-loaded) test.

Test condition	Parameter	Connect				Initial limits 25° ±5°C Ambient		Unit	Test figure
		A to pin	B to pin	C to pin	D to pin	Min	Max		
A1	CIP V_{out} (low-loaded)	15	15	Open	Open	0.1	0.8	V	9
A2	SER	16	Open	↓	16	↓	↓	↓	↓
A3	BIT 1	14	14	↓	Open	↓	↓	↓	↓
A4	BIT 2	9	9	↓	↓	↓	↓	↓	↓
A5	BIT 3	12	12	↓	↓	↓	↓	↓	↓
A6	BIT 4	2	2	↓	↓	↓	↓	↓	↓
A7	BIT 5	7	7	↓	↓	↓	↓	↓	↓
A8	BIT 6	5	5	↓	↓	↓	↓	↓	↓
A9	BIT 7	32	32	↓	↓	↓	↓	↓	↓
A10	BIT 8	34	34	↓	↓	↓	↓	↓	↓
A11	BIT 9	30	30	↓	↓	↓	↓	↓	↓
A12	BIT 10	28	28	↓	↓	↓	↓	↓	↓
A13	BIT 11	25	25	↓	↓	↓	↓	↓	↓
A14	BIT 12	24	24	Open	↓	↓	↓	↓	↓
A15	$\overline{\text{BIT 1}}$	13	Open	13	↓	↓	↓	↓	↓
A16	$\overline{\text{BIT 2}}$	8	↓	8	↓	↓	↓	↓	↓
A17	$\overline{\text{BIT 3}}$	11	↓	11	↓	↓	↓	↓	↓
A18	$\overline{\text{BIT 4}}$ V_{out} (low-loaded)	3	Open	3	Open	0.1	0.8	V	9

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TABLE IV. V_{out} (low-loaded) test - Continued.

Test condition	Parameter	Connect				Initial limits 25° ±5°C Ambient		Unit	Test figure
		A to pin	B to pin	C to pin	D to pin	Min	Max		
A19	$\overline{\text{BIT 5}}$ V_{out} (low-loaded)	6	Open	6	Open	0.1	0.8	V	9
A20	$\overline{\text{BIT 6}}$	4	↓	4	↓	↓	↓	↓	↓
A21	$\overline{\text{BIT 7}}$	31		31					
A22	$\overline{\text{BIT 8}}$	33		33					
A23	$\overline{\text{BIT 9}}$	29		29					
A24	$\overline{\text{BIT 10}}$	27		27					
A25	$\overline{\text{BIT 11}}$	26		26					
A26	$\overline{\text{BIT 12}}$	23		23					
A27	$\overline{\text{BIT 13}}$	22		22					
A28	$\overline{\text{BIT 14}}$ V_{out} (low-loaded)	21	Open	21	Open	0.1	0.8	V	9

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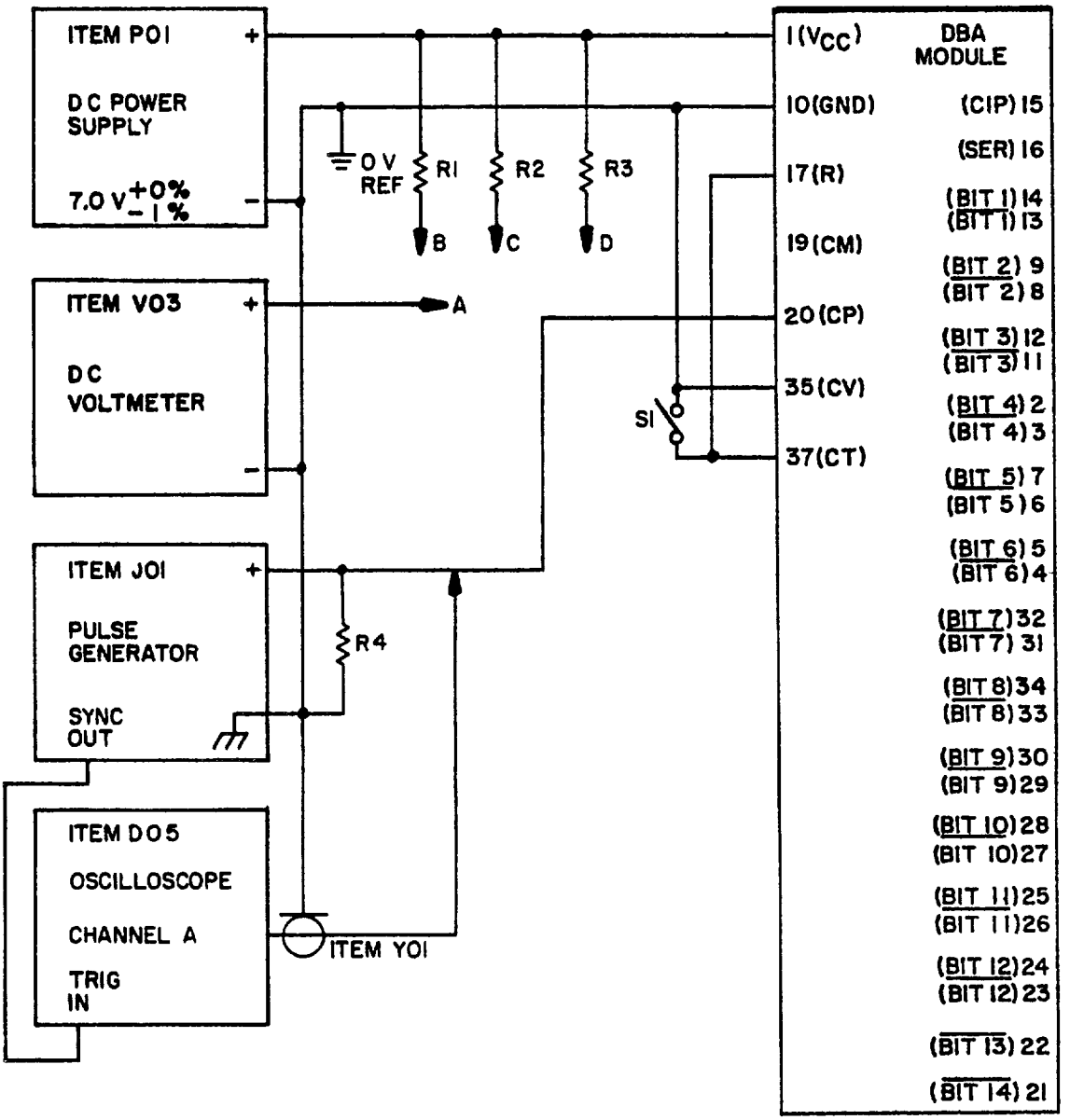


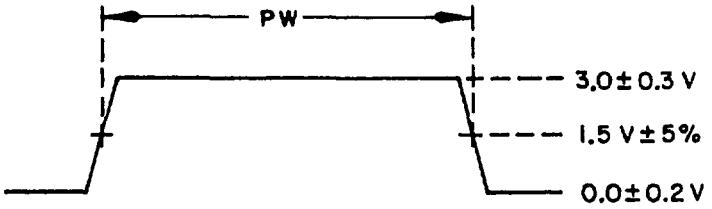
FIGURE 9. V_{out} (low-loaded) test circuit for table IV.

Source: https://assist.dla.mil -- Downloaded: 2026-10-20T16:19Z
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NOTES:
1. R1 = 634Ω ±1%, 1/8 W.
2. R2 = 536Ω ±1%, 1/8 W.
3. R3 = 316Ω ±1%, 1/8 W.
4. R4 = 51.1Ω ±5%, 1/2 W.
5. S1 = SPST switch.
6. Test procedure for test conditions A1-A14:

   a. Clock pulse parameters:

      PW = 1 ±0.1 µs.
      $t_{TLH} = t_{THL}$ = 25 ±25 ns.



   b After making all circuit connections for a test condition and with the pulse generator in the single pulse mode and switch S1 closed, apply one pulse to the circuit and make measurement. All 14 (A1-A14) measurements can be made provided power is not interrupted.

7 Test procedure for test conditions A15-A28:

   a. Same as 6a.
   b. After making all circuit connections for a test condition and with the pulse generator in the single pulse mode and switch S1 closed, apply one pulse to the circuit. Open switch S1 and apply 14 pulses to the circuit and take measurement. All 14 (A15-A28) measurements can be made provided power is not interrupted.

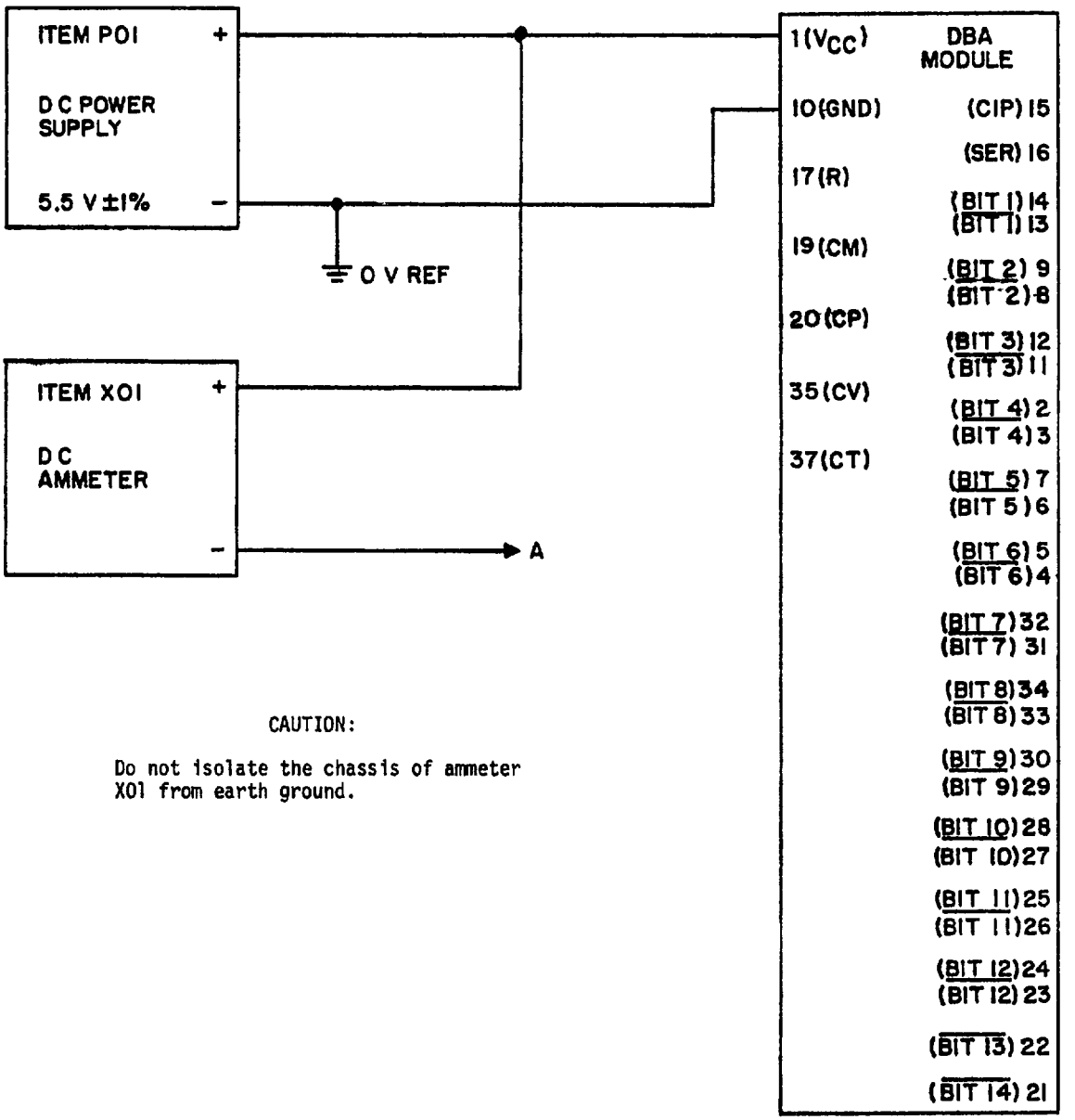
FIGURE 9. V_{out} (low-loaded) test circuit for table IV - Continued.

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TABLE V. Input breakdown current test.

Test condition	Parameter	Connect	Initial limits 25° ±5°C Ambient		Unit	Test figure
		A to pin	Min	Max		
B1	R Input breakdown current	17	0.0	1.0	mA	10
B2	CM	19		1.0		
B3	CP	20		2.0		
B4	CV	35		2.0		
B5	CT Input breakdown current	37	0.0	2.0	mA	10

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CAUTION:

Do not isolate the chassis of ammeter X01 from earth ground.

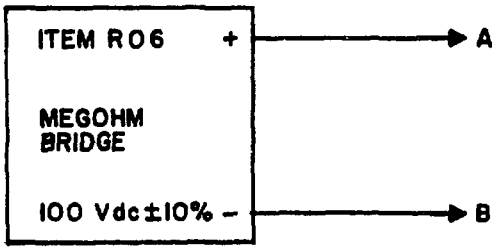
FIGURE 10. Input breakdown current test circuit for table V.

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TABLE VI. Isolation test.

Test condition	Parameter	Connect		Initial limits 25° ±5°C Ambient		Unit	Test figure
		A to pin	B to pin	Min	Max		
C1	Isolation, used to unused pins	All pins except 18,36, 38,39,40	18,36,38,39, 40	10.0	∞	MΩ	11
C2	Isolation, all pins to frame	All pins	Frame	10.0	∞	MΩ	11

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1(V_{CC})	DBA MODULE
10(GND)	(CIP) 15
	(SER) 16
17(R)	
	(BIT 1) 14
	($\overline{\text{BIT 1}}$) 13
19(CM)	
	(BIT 2) 9
	($\overline{\text{BIT 2}}$) 8
20(CP)	
	(BIT 3) 12
	($\overline{\text{BIT 3}}$) 11
35(CV)	
	(BIT 4) 2
	($\overline{\text{BIT 4}}$) 3
37(CT)	
	(BIT 5) 7
	($\overline{\text{BIT 5}}$) 6
	(BIT 6) 5
	($\overline{\text{BIT 6}}$) 4
	(BIT 7) 32
	($\overline{\text{BIT 7}}$) 31
	(BIT 8) 34
	($\overline{\text{BIT 8}}$) 33
	(BIT 9) 30
	($\overline{\text{BIT 9}}$) 29
	(BIT 10) 28
	($\overline{\text{BIT 10}}$) 27
	(BIT 11) 25
	($\overline{\text{BIT 11}}$) 26
	(BIT 12) 24
	($\overline{\text{BIT 12}}$) 23
	($\overline{\text{BIT 13}}$) 22
	($\overline{\text{BIT 14}}$) 21

WARNING:

When using a megohm bridge, high voltage may be present at any of the output binding posts. Always set the megohm bridge function switch to "discharge" before connecting or disconnecting the test leads.

NOTE: On test condition C2 (when measuring to frame), penetrate the anodized finish of the module frame with a sharply pointed probe connected to the negative terminal.

FIGURE 11. Isolation test circuit for table VI.

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TABLE VII. Input current (low-level and high-level) and output short circuit current.

Test condition	Parameter	Connect				Initial limits 25° ±5°C Ambient		Unit	Test figure
		A to pin	B to pin	C to pin	Pin 10 to pin	Min	Max		
D1	R Input current, low level	Open	17	Open	Open	0.01	5.19	mA	12
D2	CP		20			0.1	3.46		
D3	CV		35				3.46		
D4	CT		37				3.46		
D5	CM Input current, low level	Open	19		Open	0.1	1.73	mA	
E1	R Input current, high level	17	Open		35,37	0	120	µA	
E2	CP	20			Open		80		
E3	CV	35			17		80		
E4	CT	37			Open		80		
E5	CM Input current, high level	19	Open	Open	17	0	40	µA	
F1	CIP Output short circuit current	Open	15	37	Open	1	100	mA	
F2	SER		16	20	Open	1			
F3	$\overline{\text{BIT 1}}$		13		17,35, 37	10			
F4	$\overline{\text{BIT 2}}$		8						
F5	$\overline{\text{BIT 3}}$		11						
F6	$\overline{\text{BIT 4}}$		3						
F7	$\overline{\text{BIT 5}}$		6						
F8	$\overline{\text{BIT 6}}$ Output short circuit current	Open	4	20	17,35, 37	10	100	mA	12

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TABLE VII. Input current (low-level and high-level) and output short circuit current test - Continued.

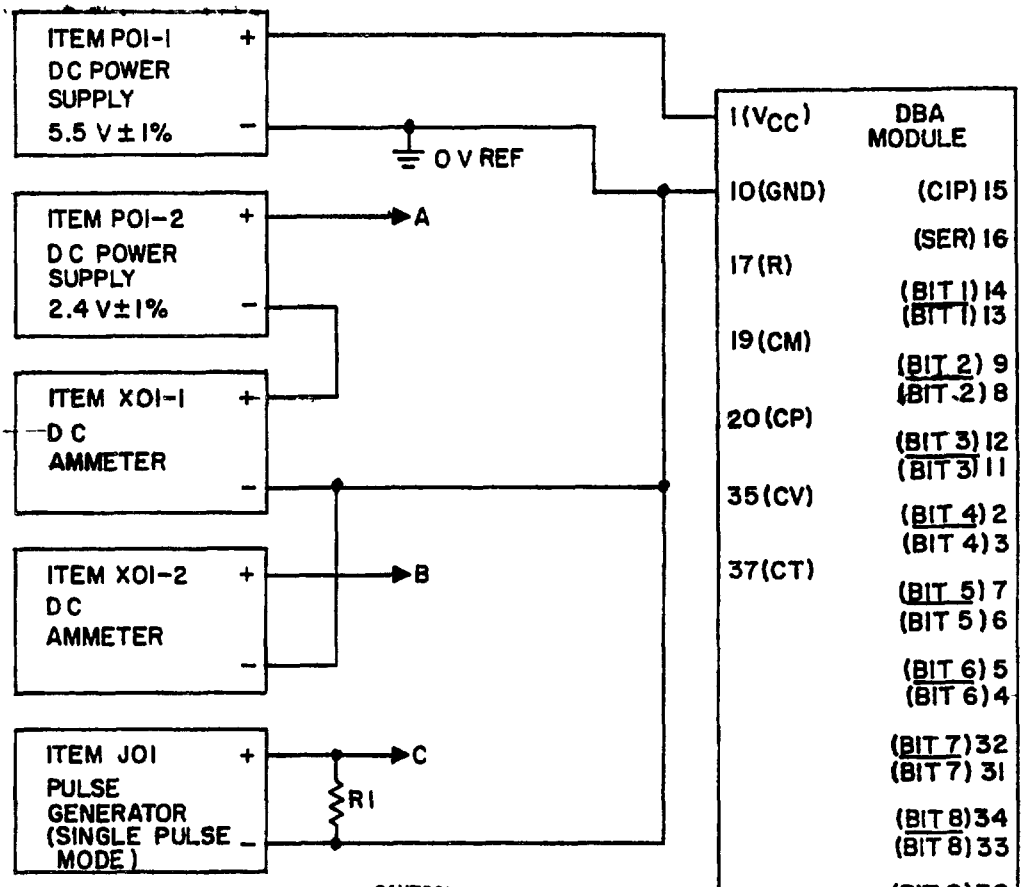
Test condition	Parameter		Connect				Initial limits 25° ±5°C Ambient		Unit	Test figure
			A to pin	B to pin	C to pin	Pin 10 to pin	Min	Max		
F9	$\overline{\text{BIT 7}}$	Output short circuit current	Open	31	20	17,35, 37	10	100	mA	12
F10	$\overline{\text{BIT 8}}$			33						
F11	$\overline{\text{BIT 9}}$			29						
F12	$\overline{\text{BIT 10}}$			27						
F13	$\overline{\text{BIT 11}}$			26						
F14	$\overline{\text{BIT 12}}$			23						
F15	$\overline{\text{BIT 13}}$			22						
F16	$\overline{\text{BIT 14}}$		Open	21		17,35, 37	10			
F17	BIT 1		17,35, 37	14		Open	20			
F18	BIT 2			9						
F19	BIT 3			12						
F20	BIT 4			2						
F21	BIT 5			7						
F22	BIT 6			5						
F23	BIT 7			32						
F24	BIT 8			34						
F25	BIT 9	Output short circuit current	17,35, 37	30	20	Open	20	100	mA	12

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TABLE VII. Input current (low-level and high-level) and output short circuit current test - Continued.

Test condition	Parameter	Connect				Initial limits 25° ±5°C Ambient		Unit	Test figure
		A to pin	B to pin	C to pin	Pin 10 to pin	Min	Max		
F26	BIT 10 Output short circuit current	17,35, 37	28	20	Open	20	100	mA	12
F27	BIT 11 Output short circuit current	17,35, 37	25	20	Open	20	100	mA	12
F28	BIT 12 Output short circuit current	17,35, 37	24	20	Open	20	100	mA	12

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CAUTION

Do not isolate the chassis of ammeter X01 from earth ground.

NOTES.
1. R1 = 51.1Ω ±5%, 1/2 W.
2. Since items X01-1 and X01-2 are not used simultaneously, only one item X01 is required.
3. Adjust item J01 in the continuous pulse mode for the following parameters:
   a. PW = 1 ±0.1 µs.
   b. $t_{TLH} = t_{THL}$ = 25 ±20 ns.
   c. Amplitude = 3 ±0.3 V.
4. Only one output shall be shorted at a time and the measurement taken within 10 seconds.
5. Test procedure for test conditions F1-F28:
   a. After making all circuit connections for each test condition and with item J01 in the single pulse mode, apply one pulse to the circuit.
   b. The power must not be interrupted during test conditions F1-F28.

FIGURE 12. Input current (low-level and high-level) and output short circuit current test circuit for table VII.

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TABLE VIII. Output voltage (input threshold conditions), low-level and high-level, test.

Test condition	Parameter		Connect		Initial limits 25° ±5°C Ambient		Unit	Test figure
			A to pin	B to pin	Min	Max		
G1	BIT 1	Output voltage, high level	35	14	2.4	4.5	V	13
G2	BIT 2			9				
G3	BIT 3			12				
G4	BIT 4			2				
G5	BIT 5			7				
G6	BIT 6			5				
G7	BIT 7			32				
G8	BIT 8			34				
G9	BIT 9			30				
G10	BIT 10			28				
G11	BIT 11			25				
G12	BIT 12			24				
G13	SER			16				
G14	CIP	Output voltage, high level		15	2.4	4.5		
H1	$\overline{\text{BIT 1}}$	$\overline{\text{Output}}$ voltage, low level		13	0.0	0.4		
H2	$\overline{\text{BIT 2}}$			8				
H3	$\overline{\text{BIT 3}}$			11				
H4	$\overline{\text{BIT 4}}$			3				
H5	$\overline{\text{BIT 5}}$	$\overline{\text{Output}}$ voltage, low level	35	6	0.0	0.4	V	13

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TABLE VIII. Output voltage (input threshold conditions), low-level and high-level, test - Continued.

Test condition	Parameter		Connect		Initial limits 25° ±5°C Ambient		Unit	Test figure
			A to pin	B to pin	Min	Max		
H6	$\overline{\text{BIT 6}}$	$\overline{\text{Output}}$ voltage, low level	35	4	0.0	0.4	V	13
H7	$\overline{\text{BIT 7}}$			31				
H8	$\overline{\text{BIT 8}}$			33				
H9	$\overline{\text{BIT 9}}$			29				
H10	$\overline{\text{BIT 10}}$			27				
H11	$\overline{\text{BIT 11}}$			26				
H12	$\overline{\text{BIT 12}}$			23				
H13	$\overline{\text{BIT 13}}$			22				
H14	$\overline{\text{BIT 14}}$	$\overline{\text{Output}}$ voltage, low level	35	21				
J1	BIT1	Output voltage, low level	35,37,17	14				
J2	BIT 2			9				
J3	BIT 3			12				
J4	BIT 4			2				
J5	BIT 5			7				
J6	BIT 6			5				
J7	BIT 7			32				
J8	BIT 8			34				
J9	BIT 9			30				
J10	BIT 10	Output voltage, low level	35,37,17	28	0.0	0.4	V	13

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TABLE VIII. Output voltage (input threshold conditions), low-level and high-level, test - Continued.

Test condition	Parameter		Connect		Initial limits 25° ±5°C Ambient		Unit	Test figure
			A to pin	B to pin	Min	Max		
J11	BIT 11	Output voltage, low level	35,37,17	25	0.0	0.4	V	13
J12	BIT 12	↓	↓	24	↓	↓	↓	↓
J13	SER	↓	↓	16	↓	↓	↓	↓
J14	CIP	Output voltage, low level	↓	15	0.0	0.4	↓	↓
K1	$\overline{\text{BIT 1}}$	$\overline{\text{Output}}$ voltage, high level	↓	13	2.4	4.5	↓	↓
K2	$\overline{\text{BIT 2}}$	↓	↓	8	↓	↓	↓	↓
K3	$\overline{\text{BIT 3}}$	↓	↓	11	↓	↓	↓	↓
K4	$\overline{\text{BIT 4}}$	↓	↓	3	↓	↓	↓	↓
K5	$\overline{\text{BIT 5}}$	↓	↓	6	↓	↓	↓	↓
K6	$\overline{\text{BIT 6}}$	↓	↓	4	↓	↓	↓	↓
K7	$\overline{\text{BIT 7}}$	↓	↓	31	↓	↓	↓	↓
K8	$\overline{\text{BIT 8}}$	↓	↓	33	↓	↓	↓	↓
K9	$\overline{\text{BIT 9}}$	↓	↓	29	↓	↓	↓	↓
K10	$\overline{\text{BIT 10}}$	↓	↓	27	↓	↓	↓	↓
K11	$\overline{\text{BIT 11}}$	↓	↓	26	↓	↓	↓	↓
K12	$\overline{\text{BIT 12}}$	↓	↓	23	↓	↓	↓	↓
K13	$\overline{\text{BIT 13}}$	↓	↓	22	↓	↓	↓	↓
K14	$\overline{\text{BIT 14}}$	$\overline{\text{Output}}$ voltage, high level	35,37,17	21	2.4	4.5	V	13

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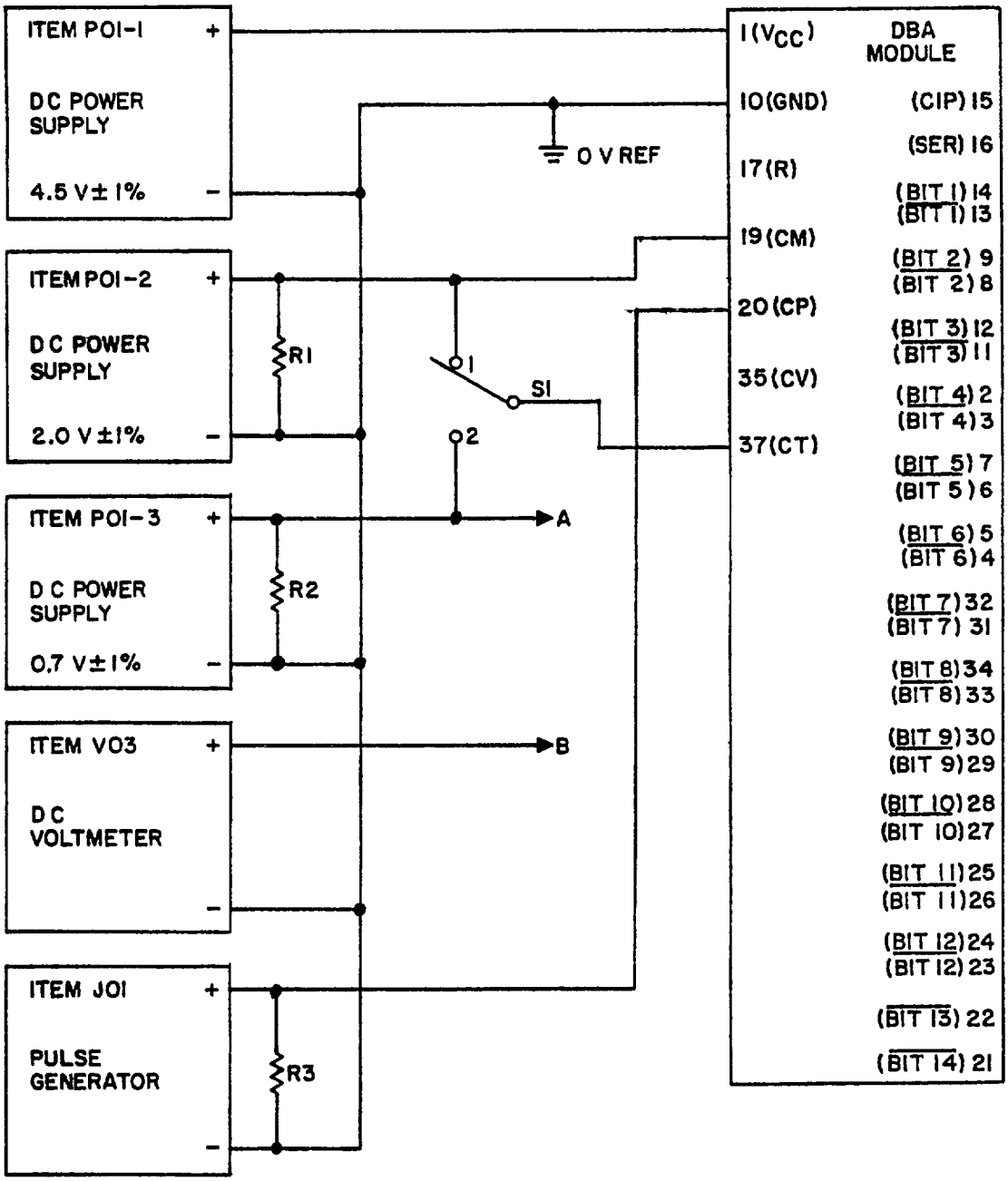
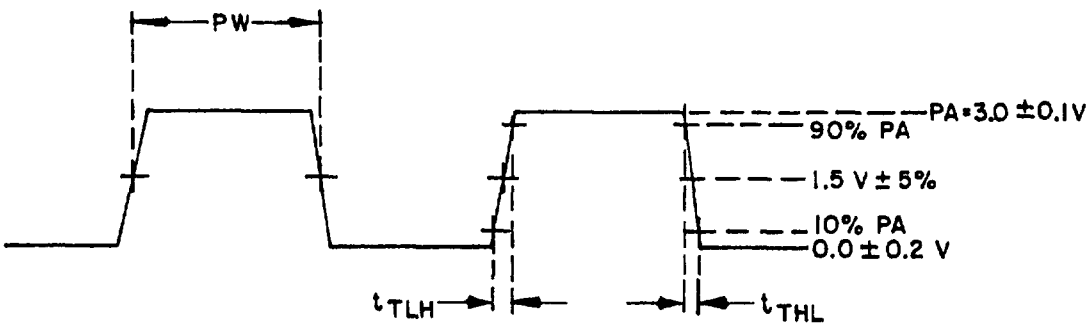


FIGURE 13. Output voltage (input threshold conditions), low-level and high-level, test circuit for table VIII

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NOTES:
1. R1 = R2 = R3 = 51.1Ω ±5%, 1/2 W.
2. S1 = SPDT switch.
3. Load all outputs as specified on figure 14.
4. Test procedure for test conditions G1-G14 and H1-H14.

   a. Clock pulse.



   PW = 1 ±0.1 µs
   $t_{TLH} = t_{THL}$ = 25 ±10 ns.

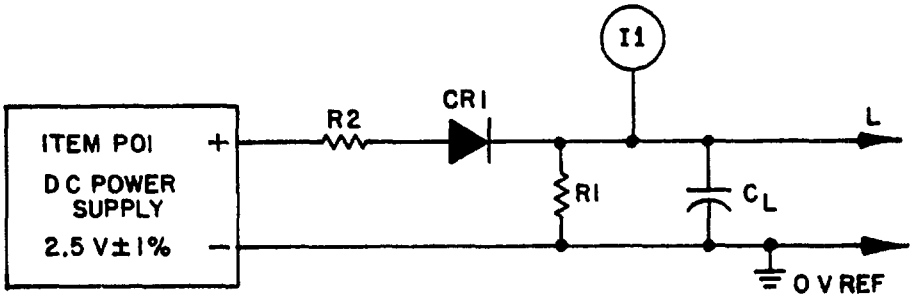
   b. With pulse generator J01 in the single pulse mode and all connections made for a test condition, put switch S1 in position 2 and pulse item J01 once, then put switch S1 in position 1 and pulse item J01 14 times and make measurements G1-G14 and H1-H14

5. Test procedure for test conditions J1-J14 and K1-K14

   With pulse generator J01 in the single pulse mode and all connections made for a test condition pulse item J01 once and make measurements J1-J14 and K1-K14

FIGURE 13 Output voltage (input threshold conditions), low-level and high-level, test circuit for table VIII - Continued.

Source: https://assist.dla.mil -- Downloaded: 2026-10-20T16:19Z
Check the source to verify that this is the current version before use.



NOTES:
1. CR1 = 1N4150, or equivalent.
2. Capacitor C_L is not required for dc electrical test conditions G, H, J, and K.
3. During output switching the 2.5 V power supply shall have a maximum overshoot of 500 mV peak.
4. Load for the serial output (SER, pin 16):
   a. R1 = 6 04 kΩ ±1%, 1/8 W.
   b. R2 = Select a value for a current flow of 16.0 mA ±1% with point L held at 0.4 volt.
   c. C_L = 50 ±5 pF (including probe and parasitic capacitance).
5. Load for the conversion in process output (CIP, pin 15)·
   a R1 = 23 7 kΩ ±1%, 1/8 W.
   b. R2 = Select a value for a current flow of 8.0 mA ±1% with point L held at 0 4 volt.
   c. C_L = 50 ±5 pF (including probe and parasitic capacitance).
6. Load for the parallel active high $\overline{\text{BIT}}$ outputs (pins 3, 4, 6, 8, 11, 13, 21, 22, 23, 26, 27, 29, 31, and 33).
   a. R1 = 4.99 kΩ ±1%, 1/8 W.
   b R2 = Select a value for a current flow of 9.6 mA ±1% with point L held at 0.4 volt.
   c. C_L = 50 ±5 pF (including probe and parasitic capacitance).
7. Load for the parallel active low BIT outputs (pins 2, 5, 7, 9, 12, 14, 24, 25, 28, 30, 32, and 34):
   a. R1 = 6.04 kΩ ±1%, 1/8 W
   b R2 = Select a value for a current flow of 8.0 mA ±1% with point L held at 0 4 volt
   c. C_L = 50 ±5 pF (including probe and parasitic capacitance).
8. I1 = a logic state indicator compatible with logic levels and drive capabilities of the DBA module (required only for test conditions M, N, and P).

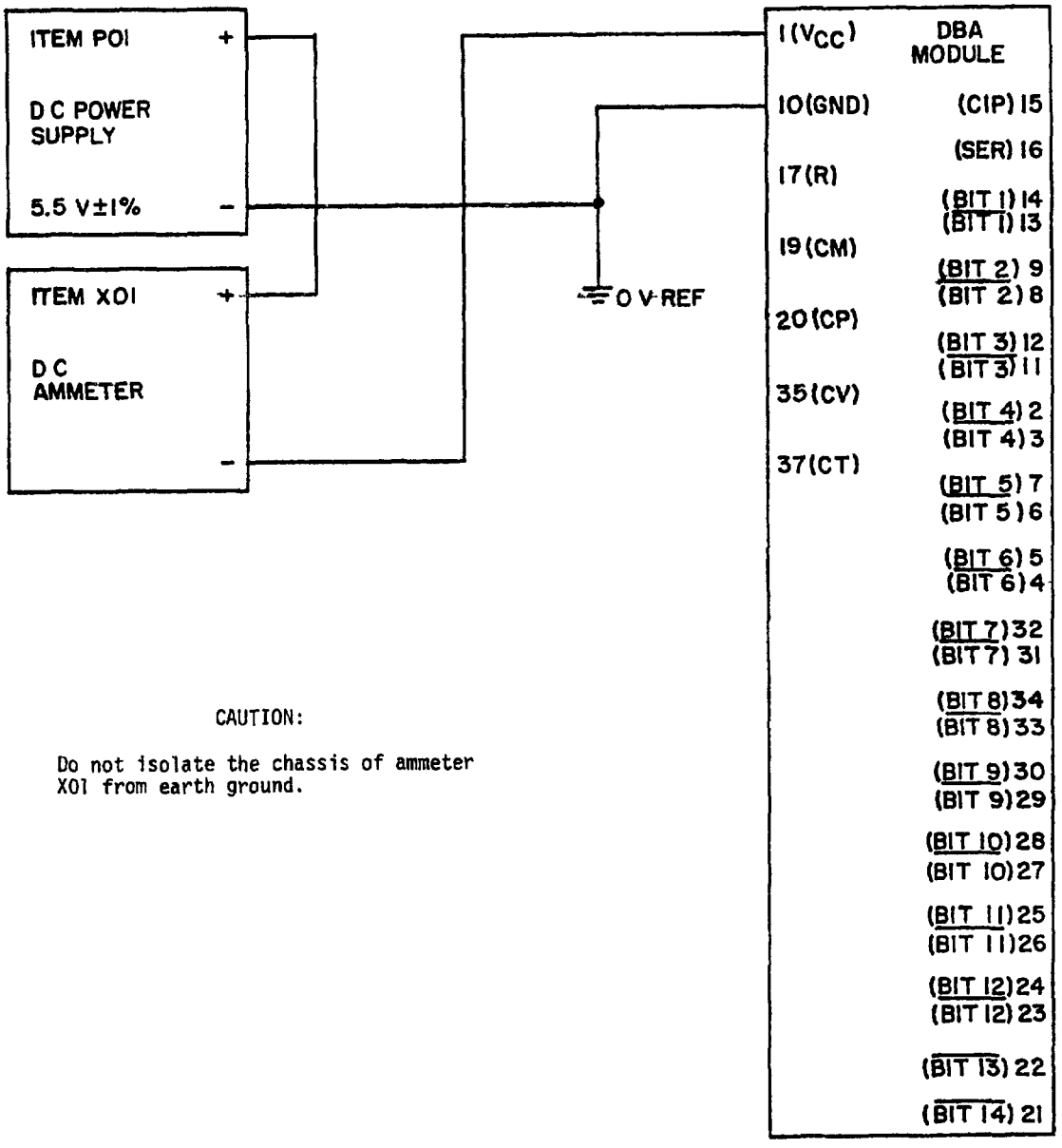
FIGURE 14. Logic load circuits.

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TABLE IX. Power supply current test.

Test condition	Parameter	Initial limits 25° ±5°C Ambient		Unit	Test figure
		Min	Max		
L	Power supply current	100	385	mA	15

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CAUTION:

Do not isolate the chassis of ammeter X01 from earth ground.

FIGURE 15. Power supply current test circuit for table IX.

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TABLE X Functional A/D conversion test

Test cond.	Parameter	Connect				Outputs																												Test figure
		Pin 17 to pin	A to pin	B to pin	No. of pulses	BIT 1	BIT 2	BIT 3	BIT 4	BIT 5	BIT 6	BIT 7	BIT 8	BIT 9	BIT 10	BIT 11	BIT 12	BIT 1	BIT 2	BIT 3	BIT 4	BIT 5	BIT 6	BIT 7	BIT 8	BIT 9	BIT 10	BIT 11	BIT 12	BIT 13	BIT 14	SER	CIP	
M1	DC convert	10	35, 19	37	1	L	L	L	L	L	L	L	L	L	L	L	L	H	H	H	H	H	H	H	H	H	H	H	H	H	H	L	L	16
M2		21	37	35, 19	1	H	L	L	L	L	L	L	L	L	L	L	L	L	H	H	H	H	H	H	H	H	H	H	H	H	L	H	
M3					1	L	H	L	L	L	L	L	L	L	L	L	L	H	L	H	H	H	H	H	H	H	H	H	H	H	L	H	
M4					1	L	L	H	L	L	L	L	L	L	L	L	L	H	H	L	H	H	H	H	H	H	H	H	H	H	L	H	
M5					1	L	L	L	H	L	L	L	L	L	L	L	L	H	H	H	L	H	H	H	H	H	H	H	H	H	L	H	
M6					1	L	L	L	L	H	L	L	L	L	L	L	L	H	H	H	H	L	H	H	H	H	H	H	H	H	L	H	
M7					1	L	L	L	L	L	H	L	L	L	L	L	L	H	H	H	H	H	L	H	H	H	H	H	H	H	L	H	
M8					1	L	L	L	L	L	L	H	L	L	L	L	L	H	H	H	H	H	H	L	H	H	H	H	H	H	L	H	
M9					1	L	L	L	L	L	L	L	H	L	L	L	L	H	H	H	H	H	H	H	L	H	H	H	H	H	L	H	
M10					1	L	L	L	L	L	L	L	L	H	L	L	L	H	H	H	H	H	H	H	H	L	H	H	H	H	L	H	
M11					1	L	L	L	L	L	L	L	L	L	H	L	L	H	H	H	H	H	H	H	H	H	L	H	H	H	L	H	
M12					1	L	L	L	L	L	L	L	L	L	L	H	L	H	H	H	H	H	H	H	H	H	H	L	H	H	L	H	
M13					1	L	L	L	L	L	L	L	L	L	L	L	H	H	H	H	H	H	H	H	H	H	H	H	L	H	L	H	
M14			↓	↓	1	L	L	L	L	L	L	L	L	L	L	L	L	H	H	H	H	H	H	H	H	H	H	H	H	L	H	L	H	
M15			37	35, 19	1	L	L	L	L	L	L	L	L	L	L	L	L	H	H	H	H	H	H	H	H	H	H	H	H	L	L	L	L	
M16			19	35, 37	1	L	L	L	L	L	L	L	L	L	L	L	L	H	H	H	H	H	H	H	H	H	H	H	H	L	L	L	L	
M17			37, 19	35	1	H	L	L	L	L	L	L	L	L	L	L	L	L	H	H	H	H	H	H	H	H	H	H	H	L	L	H		
M18					1	H	H	L	L	L	L	L	L	L	L	L	L	L	L	H	H	H	H	H	H	H	H	H	H	H	H	H		
M19					1	H	H	H	L	L	L	L	L	L	L	L	L	L	L	L	H	H	H	H	H	H	H	H	H	H	H	H		
M20					1	H	H	H	H	L	L	L	L	L	L	L	L	L	L	L	L	H	H	H	H	H	H	H	H	H	H	H		
M21					1	H	H	H	H	H	L	L	L	L	L	L	L	L	L	L	L	L	H	H	H	H	H	H	H	H	H	H		
M22					1	H	H	H	H	H	H	L	L	L	L	L	L	L	L	L	L	L	L	H	H	H	H	H	H	H	H	H		
M23					1	H	H	H	H	H	H	H	L	L	L	L	L	L	L	L	L	L	L	L	H	H	H	H	H	H	H	H		
M24					1	H	H	H	H	H	H	H	H	L	L	L	L	L	L	L	L	L	L	L	L	H	H	H	H	H	H	H		
M25					1	H	H	H	H	H	H	H	H	H	L	L	L	L	L	L	L	L	L	L	L	L	H	H	H	H	H	H		
M26					1	H	H	H	H	H	H	H	H	H	H	L	L	L	L	L	L	L	L	L	L	L	L	H	H	H	H	H		
M27					1	H	H	H	H	H	H	H	H	H	H	H	L	L	L	L	L	L	L	L	L	L	L	L	H	H	H	H		
M28					1	H	H	H	H	H	H	H	H	H	H	H	H	L	L	L	L	L	L	L	L	L	L	L	L	H	H	H		
M29					1	H	H	H	H	H	H	H	H	H	H	H	H	L	L	L	L	L	L	L	L	L	L	L	L	H	H	H		
M30			↓	↓	1	H	H	H	H	H	H	H	H	H	H	H	H	L	L	L	L	L	L	L	L	L	L	L	L	L	L	L		
M31			37, 19	35	4	H	H	H	H	H	H	H	H	H	H	H	H	L	L	L	L	L	L	L	L	L	L	L	L	X	L	L		
M32	↓	↓	Open	37,35,19	1	L	L	L	L	L	L	L	L	L	L	L	L	H	H	H	H	H	H	H	H	H	H	H	H	H	L	L		
M33	DC convert	21	37	35, 19	1	H	L	L	L	L	L	L	L	L	L	L	L	H	H	H	H	H	H	H	H	H	H	H	H	H	L	H		16

See notes at end of table.

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TABLE X. Functional A/D conversion test - Continued.

Test cond.	Parameter	Connect				Outputs																												Test figure
		Pin 17 to pin	A to pin	B to pin	No. of pulses	BIT 1	BIT 2	BIT 3	BIT 4	BIT 5	BIT 6	BIT 7	BIT 8	BIT 9	BIT 10	BIT 11	BIT 12	BIT 1	BIT 2	BIT 3	BIT 4	BIT 5	BIT 6	BIT 7	BIT 8	BIT 9	BIT 10	BIT 11	BIT 12	BIT 13	BIT 14	SER	CIP	
N1	AC convert	10	35, 37	19	1	L	L	L	L	L	L	L	L	L	L	L	L	H	H	H	H	H	H	H	H	H	H	H	H	H	H	L	L	16
N2		21	35, 37	19	1	H	L	L	L	L	L	L	L	L	L	L	L	L	H	H	H	H	H	H	H	H	H	H	H	H	H	L	H	
N3			35	19, 37	1	L	H	L	L	L	L	L	L	L	L	L	L	H	L	H	H	H	H	H	H	H	H	H	H	H	L	H	
N4					1	L	L	H	L	L	L	L	L	L	L	L	L	H	H	L	H	H	H	H	H	H	H	H	H	H	L	H	
N5					1	L	L	L	H	L	L	L	L	L	L	L	L	H	H	H	L	H	H	H	H	H	H	H	H	H	L	H	
N6					1	L	L	L	L	H	L	L	L	L	L	L	L	H	H	H	H	L	H	H	H	H	H	H	H	H	L	H	
N7					1	L	L	L	L	L	H	L	L	L	L	L	L	H	H	H	H	H	L	H	H	H	H	H	H	H	L	H	
N8					1	L	L	L	L	L	L	H	L	L	L	L	L	H	H	H	H	H	H	L	H	H	H	H	H	H	L	H	
N9					1	L	L	L	L	L	L	L	H	L	L	L	L	H	H	H	H	H	H	H	L	H	H	H	H	H	L	H	
N10					1	L	L	L	L	L	L	L	L	H	L	L	L	H	H	H	H	H	H	H	H	L	H	H	H	H	L	H	
N11					1	L	L	L	L	L	L	L	L	L	H	L	L	H	H	H	H	H	H	H	H	H	L	H	H	H	L	H	
N12					1	L	L	L	L	L	L	L	L	L	L	H	L	H	H	H	H	H	H	H	H	H	H	L	H	H	L	H	
N13					1	L	L	L	L	L	L	L	L	L	L	L	H	H	H	H	H	H	H	H	H	H	H	H	L	H	H	L	H	
N14					1	L	L	L	L	L	L	L	L	L	L	L	L	H	H	H	H	H	H	H	H	H	H	H	H	H	L	H	
N15			35	19, 37	2	L	L	L	L	L	L	L	L	L	L	L	L	H	H	H	H	H	H	H	H	H	H	H	H	H	L	L	
N16			35, 37	19	1	H	L	L	L	L	L	L	L	L	L	L	L	L	H	H	H	H	H	H	H	H	H	H	H	H	L	H	
N17			35, 19	37	1	H	H	L	L	L	L	L	L	L	L	L	L	L	L	H	H	H	H	H	H	H	H	H	H	H	H	H	
N18					1	H	H	H	L	L	L	L	L	L	L	L	L	L	L	L	H	H	H	H	H	H	H	H	H	H	H	H	
N19					1	H	H	H	H	L	L	L	L	L	L	L	L	L	L	L	L	H	H	H	H	H	H	H	H	H	H	H	
N20					1	H	H	H	H	H	L	L	L	L	L	L	L	L	L	L	L	L	H	H	H	H	H	H	H	H	H	H	
N21					1	H	H	H	H	H	H	L	L	L	L	L	L	L	L	L	L	L	L	H	H	H	H	H	H	H	H	H	
N22					1	H	H	H	H	H	H	H	L	L	L	L	L	L	L	L	L	L	L	L	H	H	H	H	H	H	H	H	
N23					1	H	H	H	H	H	H	H	H	L	L	L	L	L	L	L	L	L	L	L	L	H	H	H	H	H	H	H	
N24					1	H	H	H	H	H	H	H	H	H	L	L	L	L	L	L	L	L	L	L	L	L	H	H	H	H	H	H	
N25					1	H	H	H	H	H	H	H	H	H	H	L	L	L	L	L	L	L	L	L	L	L	L	H	H	H	H	H	
N26					1	H	H	H	H	H	H	H	H	H	H	H	L	L	L	L	L	L	L	L	L	L	L	L	H	H	H	H	
N27					1	H	H	H	H	H	H	H	H	H	H	H	H	L	L	L	L	L	L	L	L	L	L	L	L	H	H	H	
N28					1	H	H	H	H	H	H	H	H	H	H	H	H	L	L	L	L	L	L	L	L	L	L	L	L	L	H	H	
N29					2	L	L	L	L	L	L	L	L	L	L	L	L	H	H	H	H	H	H	H	H	H	H	H	H	H	L	L	
N30			35, 19	37	4	L	L	L	L	L	L	L	L	L	L	L	L	H	H	H	H	H	H	H	H	H	H	H	H	H	L	L	
N31	AC convert	21	35, 37	19	1	H	L	L	L	L	L	L	L	L	L	L	L	L	H	H	H	H	H	H	H	H	H	H	H	H	H	H	H	16

See notes at end of table.

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TABLE X. Functional A/D conversion test - Continued.

Test cond	Parameter	Connect				Outputs																												Test figure
		Pin 17 to pin	A to pin	B to pin	No of pulses	BIT 1	BIT 2	BIT 3	BIT 4	BIT 5	BIT 6	BIT 7	BIT 8	BIT 9	BIT 10	BIT 11	BIT 12	BIT 1	BIT 2	BIT 3	BIT 4	BIT 5	BIT 6	BIT 7	BIT 8	BIT 9	BIT 10	BIT 11	BIT 12	BIT 13	BIT 14	SER	CIP	
* P1	Continuous convert	10	35, 37	19	1	L	L	L	L	L	L	L	L	L	L	L	L	H	H	H	H	H	H	H	H	H	H	H	H	H	H	L	L	16
P2		21			1	H	L	L	L	L	L	L	L	L	L	L	L	L	H	H	H	H	H	H	H	H	H	H	H	H	H	L	H	
P3					1	L	H	L	L	L	L	L	L	L	L	L	L	H	L	H	H	H	H	H	H	H	H	H	H	H	H	L	H	
P4					1	L	L	H	L	L	L	L	L	L	L	L	L	H	H	L	H	H	H	H	H	H	H	H	H	H	H	L	H	
P5					1	L	L	L	H	L	L	L	L	L	L	L	L	H	H	H	L	H	H	H	H	H	H	H	H	H	H	L	H	
P6					1	L	L	L	L	H	L	L	L	L	L	L	L	H	H	H	H	L	H	H	H	H	H	H	H	H	H	L	H	
P7					1	L	L	L	L	L	H	L	L	L	L	L	L	H	H	H	H	H	L	H	H	H	H	H	H	H	H	L	H	
P8					1	L	L	L	L	L	L	H	L	L	L	L	L	H	H	H	H	H	H	L	H	H	H	H	H	H	H	L	H	
P9					1	L	L	L	L	L	L	L	H	L	L	L	L	H	H	H	H	H	H	H	L	H	H	H	H	H	H	L	H	
P10					1	L	L	L	L	L	L	L	L	H	L	L	L	H	H	H	H	H	H	H	H	L	H	H	H	H	H	L	H	
P11					1	L	L	L	L	L	L	L	L	L	H	L	L	H	H	H	H	H	H	H	H	H	L	H	H	H	H	L	H	
P12					1	L	L	L	L	L	L	L	L	L	L	H	L	H	H	H	H	H	H	H	H	H	H	L	H	H	H	L	H	
P13					1	L	L	L	L	L	L	L	L	L	L	L	H	H	H	H	H	H	H	H	H	H	H	H	L	H	H	L	H	
P14					1	L	L	L	L	L	L	L	L	L	L	L	L	H	H	H	H	H	H	H	H	H	H	H	H	L	H	H	L	H	
P15			35, 37	19	2	L	L	L	L	L	L	L	L	L	L	L	L	H	H	H	H	H	H	H	H	H	H	H	H	L	H	H	L	L
P16			35,37,19	Open	1	H	L	L	L	L	L	L	L	L	L	L	L	L	H	H	H	H	H	H	H	H	H	H	H	H	L	H	H	L	H
P17					1	H	H	L	L	L	L	L	L	L	L	L	L	L	L	H	H	H	H	H	H	H	H	H	H	H	H	H	H	H
P18					1	H	H	H	L	L	L	L	L	L	L	L	L	L	L	L	H	H	H	H	H	H	H	H	H	H	H	H	H	H
P19					1	H	H	H	H	L	L	L	L	L	L	L	L	L	L	L	L	H	H	H	H	H	H	H	H	H	H	H	H	H
P20					1	H	H	H	H	H	L	L	L	L	L	L	L	L	L	L	L	L	H	H	H	H	H	H	H	H	H	H	H	H
P21					1	H	H	H	H	H	H	L	L	L	L	L	L	L	L	L	L	L	L	H	H	H	H	H	H	H	H	H	H	H
P22					1	H	H	H	H	H	H	H	L	L	L	L	L	L	L	L	L	L	L	L	H	H	H	H	H	H	H	H	H	H
P23					1	H	H	H	H	H	H	H	H	L	L	L	L	L	L	L	L	L	L	L	L	H	H	H	H	H	H	H	H	H
P24					1	H	H	H	H	H	H	H	H	H	L	L	L	L	L	L	L	L	L	L	L	L	H	H	H	H	H	H	H	H
P25					1	H	H	H	H	H	H	H	H	H	H	L	L	L	L	L	L	L	L	L	L	L	L	H	H	H	H	H	H	H
P26					1	H	H	H	H	H	H	H	H	H	H	H	L	L	L	L	L	L	L	L	L	L	L	L	H	H	H	H	H	H
P27					1	H	H	H	H	H	H	H	H	H	H	H	H	L	L	L	L	L	L	L	L	L	L	L	L	H	H	H	H	H
P28					1	H	H	H	H	H	H	H	H	H	H	H	H	L	L	L	L	L	L	L	L	L	L	L	L	H	H	H	H	H
P29			35,37,19	Open	2	L	L	L	L	L	L	L	L	L	L	L	L	H	H	H	H	H	H	H	H	H	H	H	H	L	H	H	L	H
P30					1	H	L	L	L	L	L	L	L	L	L	L	L	L	H	H	H	H	H	H	H	H	H	H	H	H	L	H	H	L	H
P31	Continuous convert	21	35, 37	19	1	L	H	L	L	L	L	L	L	L	L	L	L	H	L	H	H	H	H	H	H	H	H	H	H	H	H	L	H	16

NOTES:
1. L = Low level.
2. H = High level.
3. X = Don't care condition.

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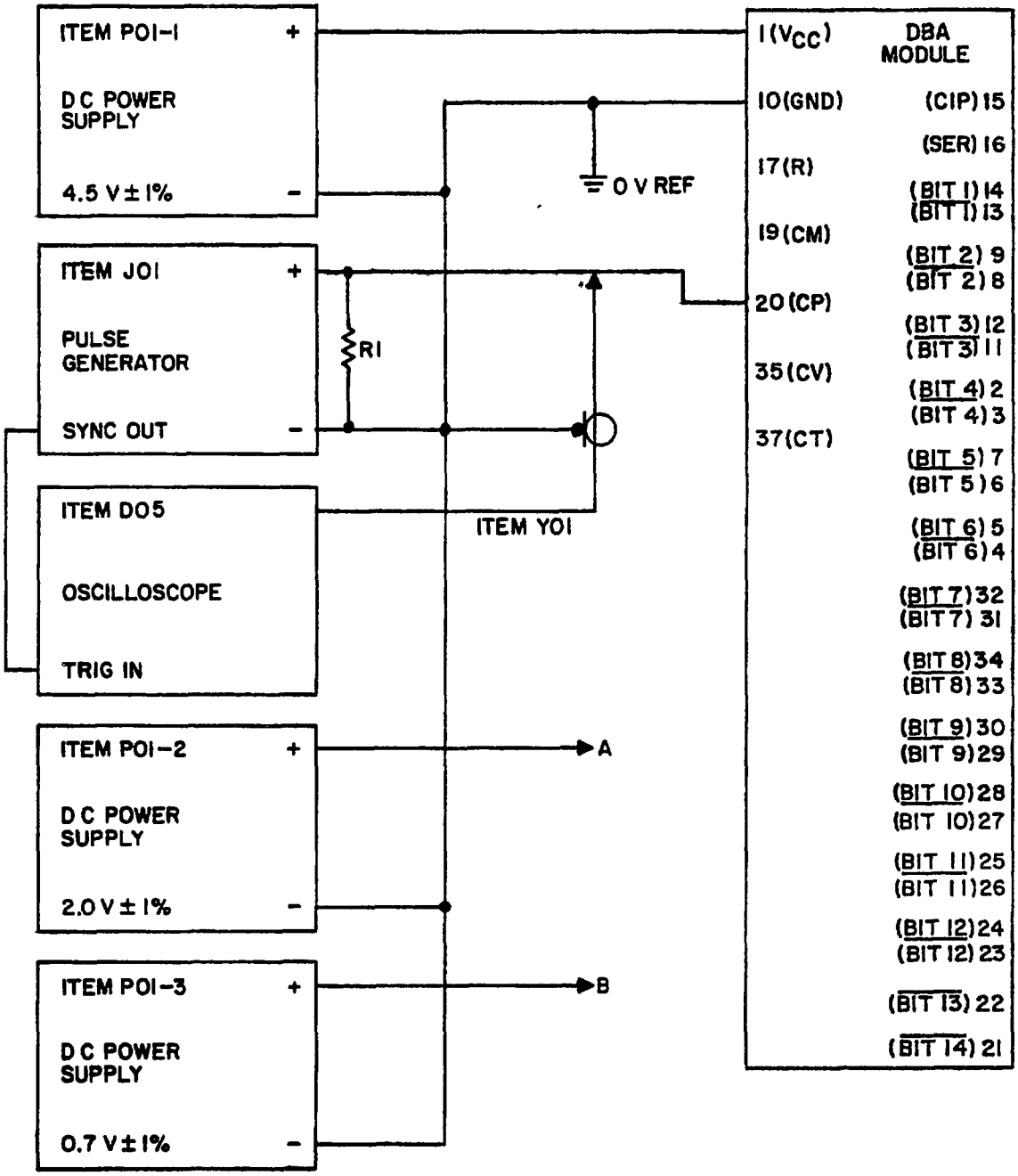
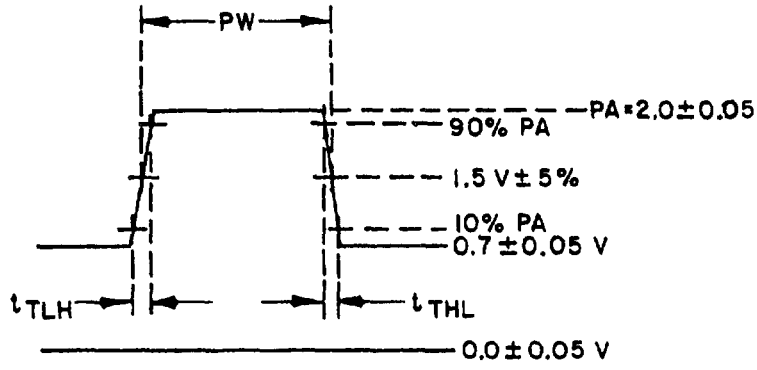


FIGURE 16. Functional A/D conversion test circuit for table X.

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NOTES.
1. R1 = 51 1Ω ±5%, 1/2 W.
2 Load all outputs under test with the load as specified on figure 14.
3. Pulse generator setup



PW = 1 ±0.1 µs.
$t_{TLH} = t_{THL}$ = 100 ±10 ns.

4 Place the pulse generator in the single pulse mode and apply the required number of pulses after all circuit connections are made, then verify that the output states are as specified in table X.

FIGURE 16. Functional A/D conversion test circuit for table X - Continued.

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TABLE XI. Output transition time (t_{TLH} and t_{THL}) test.

Test cond.	Parameter		Connect A to pin	Connect B to pin	Connect C to pin	Connect D to pin	Connect Pin 17 to pin	t_{TLH} initial limits 25° ±5°C ambient Min	t_{TLH} initial limits 25° ±5°C ambient Max	t_{THL} initial limits 25° ±5°C ambient Min	t_{THL} initial limits 25° ±5°C ambient Max	Unit	Test figure
Q1, R1	BIT 1	Output transition time	Open	Open	14	35, 37	8	5	50	2	20	ns	17, 18
Q2, R2	$\overline{\text{BIT 1}}$	↓	↓	↓	13	↓	8	↓	↓	↓	↓	↓	↓
Q3, R3	BIT 2				9		11						
Q4, R4	$\overline{\text{BIT 2}}$				8		11						
Q5, R5	BIT 3				12		3						
Q6, R6	$\overline{\text{BIT 3}}$				11		3						
Q7, R7	BIT 4				2		6						
Q8, R8	$\overline{\text{BIT 4}}$				3		6						
Q9, R9	BIT 5				7		4						
Q10, R10	$\overline{\text{BIT 5}}$				6		4						
Q11, R11	BIT 6				5		31						
Q12, R12	$\overline{\text{BIT 6}}$				4		31						
Q13, R13	BIT 7				32		33						
Q14, R14	$\overline{\text{BIT 7}}$				31		33						
Q15, R15	BIT 8				34		29						
Q16, R16	$\overline{\text{BIT 8}}$				33		29						
Q17, R17	BIT 9				30		27						
Q18, R18	$\overline{\text{BIT 9}}$				29		27						
Q19, R19	BIT 10	↓	↓	↓	28	↓	26	↓	↓	↓	↓	↓	↓
Q20, R20	$\overline{\text{BIT 10}}$	Output transition time	Open	Open	27	35, 37	26	5	50	2	20	ns	17, 18

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TABLE XI. Output transition time (t_{TLH} and t_{THL}) test - Continued.

Test cond.	Parameter		Connect					t_{TLH} initial limits 25° ±5°C ambient		t_{THL} initial limits 25° ±5°C ambient		Unit	Test figure
			A to pin	B to pin	C to pin	D to pin	Pin 17 to pin	Min	Max	Min	Max		
Q21, R21	BIT 11	Output transition time	Open	Open	25	35, 37	23	5	50	2	20	ns	17, 18
Q22, R22	$\overline{\text{BIT 11}}$	↓	↓	↓	26	↓	23	↓	↓	↓	↓	↓	↓
Q23, R23	BIT 12				24		22						
Q24, R24	$\overline{\text{BIT 12}}$				23		22						
Q25, R25	SER				16		11						
Q26, R26	CIP	Output transition time	Open	Open	15	35, 37	8	5	50	2	20	ns	17, 18

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TABLE XII. Propagation delay time (t_{PLH} and t_{PHL}) test and setup and hold time check.

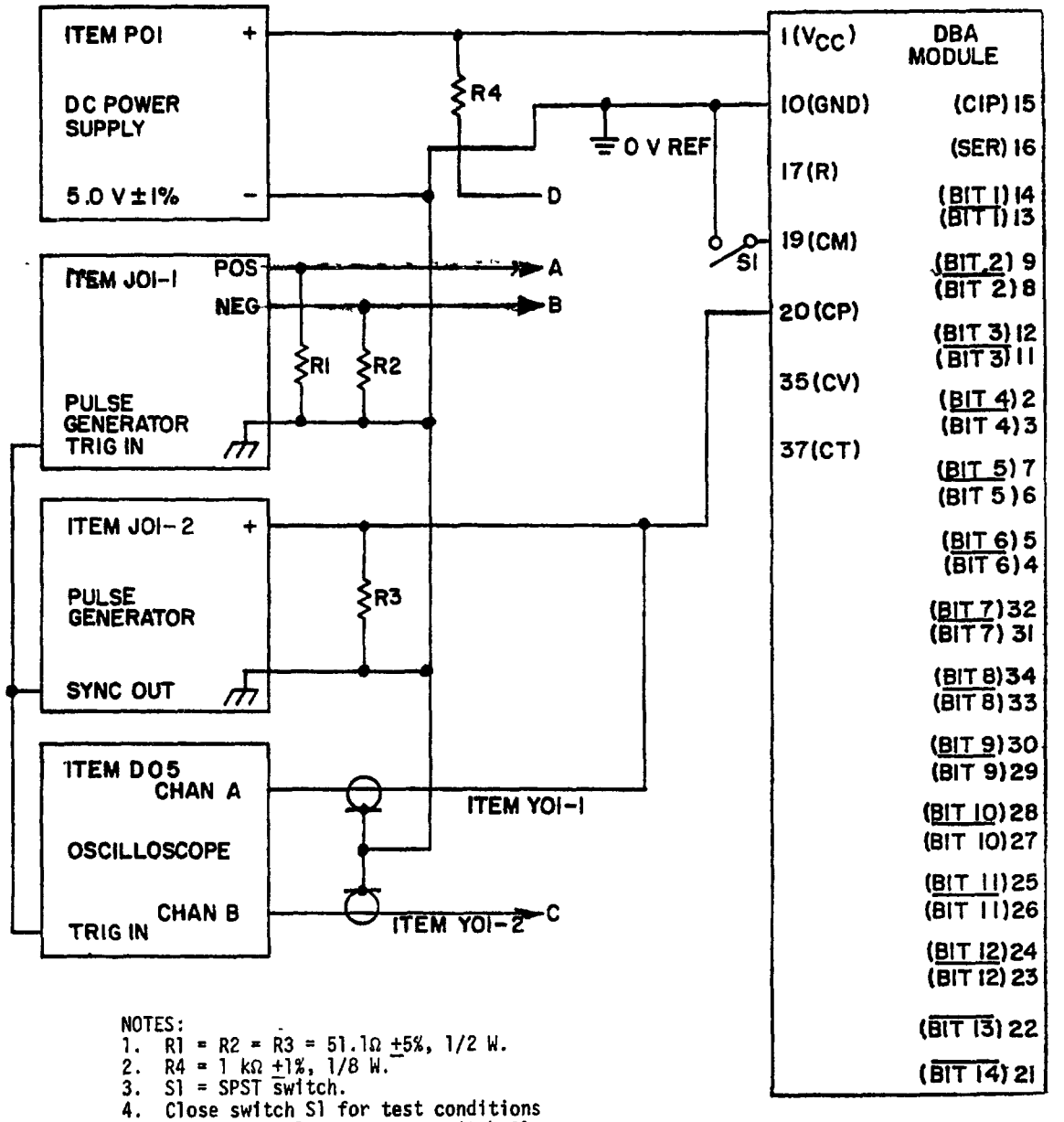
Test cond.	Parameter		Connect					t_{PLH} initial limits 25° ±5°C ambient		t_{PHL} initial limits 25° ±5°C ambient		Unit	Test figure
			A to pin	B to pin	C to pin	D to pin	Pin 17 to pin	Min	Max	Min	Max		
S1, T1	BIT 1	Propagation delay time	37	19	14	35	11	100	600	10	600	ns	17, 18
S2, T2	$\overline{\text{BIT 1}}$				13		11	10	620	100	620		
S3, T3	BIT 2				9		3	100	600	10	600		
S4, T4	$\overline{\text{BIT 2}}$				8		3	10	620	100	620		
S5, T5	BIT 3				12		6	100	600	10	600		
S6, T6	$\overline{\text{BIT 3}}$				11		6	10	620	100	620		
S7, T7	BIT 4				2		4	100	600	10	600		
S8, T8	$\overline{\text{BIT 4}}$				3		4	10	620	100	620		
S9, T9	BIT 5				7		31	100	600	10	600		
S10, T10	$\overline{\text{BIT 5}}$				6		31	10	620	100	620		
S11, T11	BIT 6				5		33	100	600	10	600		
S12, T12	$\overline{\text{BIT 6}}$				4		33	10	620	100	620		
S13, T13	BIT 7				32		29	100	600	10	600		
S14, T14	$\overline{\text{BIT 7}}$				31		29	10	620	100	620		
S15, T15	BIT 8				34		27	100	600	10	600		
S16, T16	$\overline{\text{BIT 8}}$				33		27	10	620	100	620		
S17, T17	BIT 9				30		26	100	600	10	600		
S18, T18	$\overline{\text{BIT 9}}$				29		26	10	620	100	620		
S19, T19	BIT 10				28		23	100	600	10	600		
S20, T20	$\overline{\text{BIT 10}}$	Propagation delay time	37	19	27	35	23	10	620	100	620	ns	17, 18

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TABLE XII. Propagation delay time (t_{PLH} and t_{PHL}) test and setup and hold time check - Continued.

Test cond.	Parameter	Connect A to pin	Connect B to pin	Connect C to pin	Connect D to pin	Connect Pin 17 to pin	t_{PLH} initial limits 25° ±5°C ambient Min	t_{PLH} initial limits 25° ±5°C ambient Max	t_{PHL} initial limits 25° ±5°C ambient Min	t_{PHL} initial limits 25° ±5°C ambient Max	Unit	Test figure
S21, T21	BIT 11 Propagation delay time	37	19	25	35	22	100	600	10	600	ns	17, 18
S22, T22	$\overline{\text{BIT 11}}$	↓	↓	26	↓	22	10	620	100	620	↓	↓
S23, T23	BIT 12	↓	↓	24	↓	21	100	600	10	600	↓	↓
S24, T24	$\overline{\text{BIT 12}}$	37	19	23	35	21	10	620	100	620	↓	17, 18
S25, T25	SER	19	Open	16	35, 37	21	10	100	10	115	↓	17, 19
T26	R to BIT 1	17	↓	14	↓	Open	---	---	30	250	↓	↓
S26	CP to CIP	↓	↓	15	↓	↓	30	200	---	---	↓	↓
T27	R to CIP	↓	↓	15	↓	↓	---	---	5	80	↓	17, 19
T28	R to SER Propagation delay time	17	Open	16	35, 37	Open	---	---	10	120	ns	17, 20

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NOTES:
1. R1 = R2 = R3 = 51.1Ω ±5%, 1/2 W.
2. R4 = 1 kΩ ±1%, 1/8 W.
3. S1 = SPST switch.
4. Close switch S1 for test conditions Q1-Q24 and R1-R24. Open switch S1 for test conditions Q25, Q26, R25, R26, S1-S26, and T1-T28.
5. Load all outputs except pin 21 with the load as specified on figure 14.

FIGURE 17. Output transition time (t_{TLH} and t_{THL}) and propagation delay time (t_{PLH} and t_{PHL}) test and setup and hold time check test circuit for tables XI and XII.

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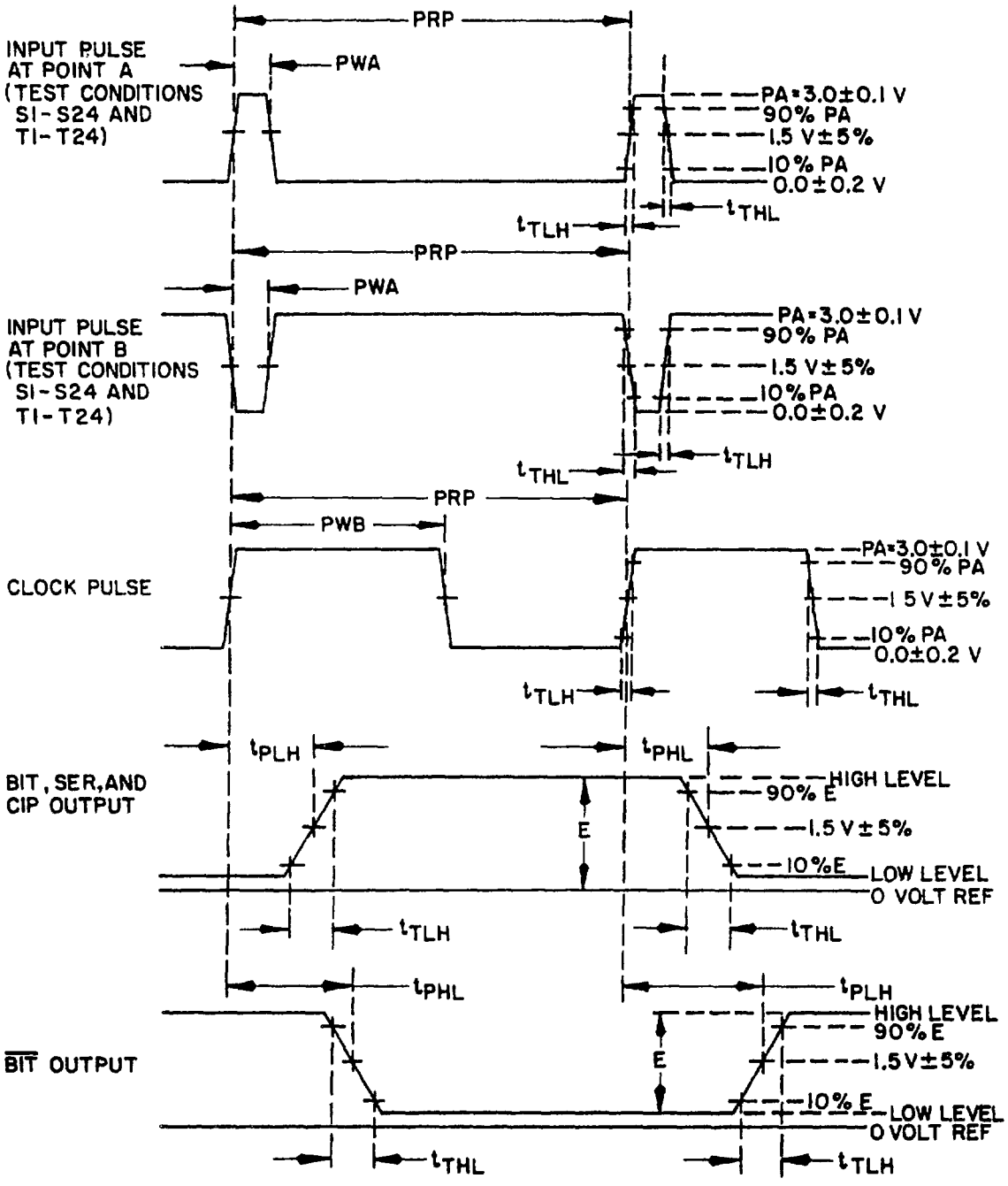


FIGURE 18. Output transition time (t_{TLH} and t_{THL}) and propagation delay time (t_{PLH} and t_{PHL}) test and setup and hold time check test circuit waveforms for tables XI and XII.

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NOTES:
1. Clock pulse parameters for test conditions Q1-Q26 and R1-R26:

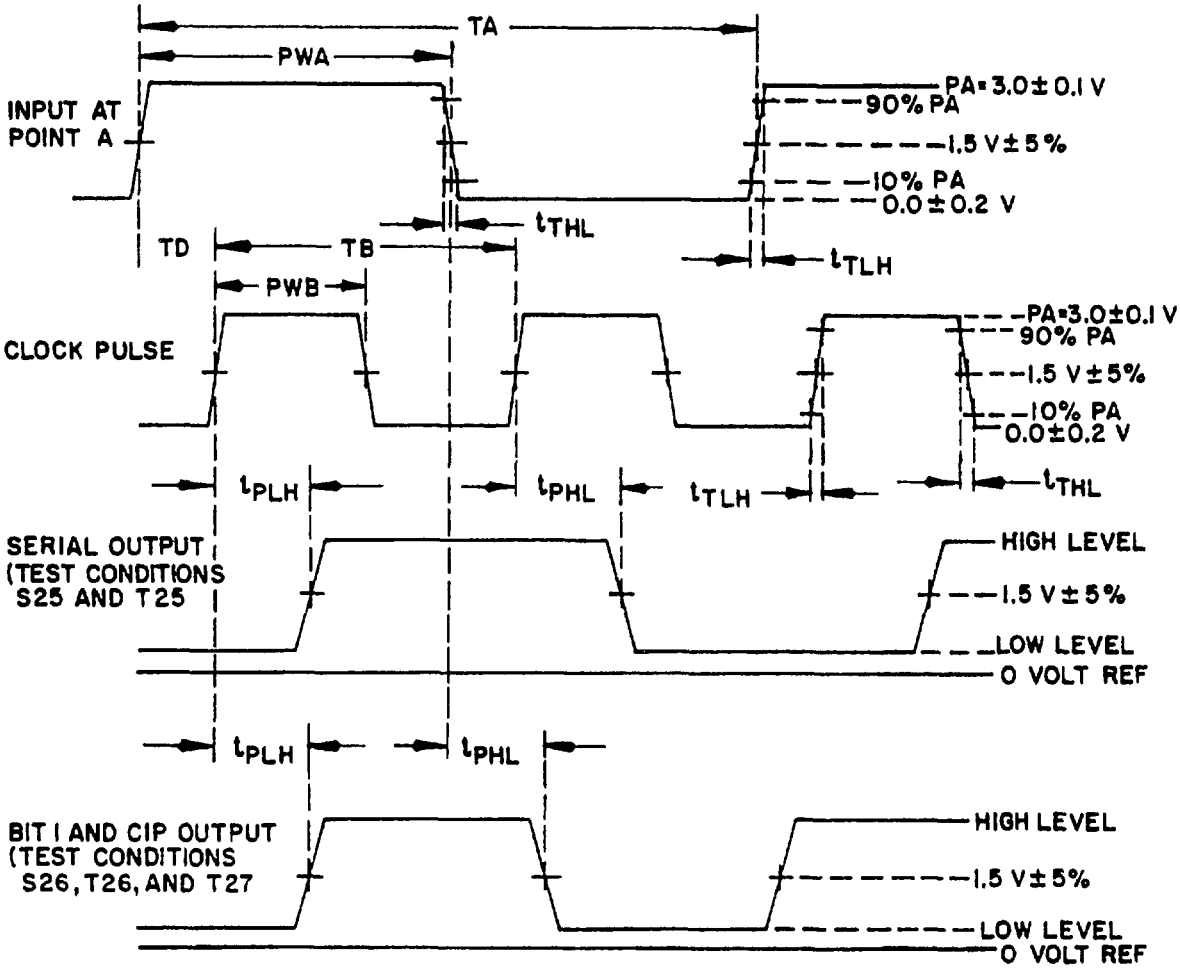
   PRP = 10 ±1 µs.
   PWB = 5 ±0.5 µs.
   $t_{TLH} = t_{THL}$ = 10 ±2 ns.

2. Input and clock pulse parameters for test conditions S1-S24 and T1-T24.

   PRP = 10 ±1 µs.
   PWA = 100 ±10 ns.
   PWB = 5 ±0.5 µs.
   $t_{TLH} = t_{THL}$ = 10 ±2 ns.

FIGURE 18. Output transition time (t_{TLH} and t_{THL}) and propagation delay time (t_{PLH} and t_{PHL}) test and setup and hold time check test circuit waveforms for tables XI and XII - Continued.

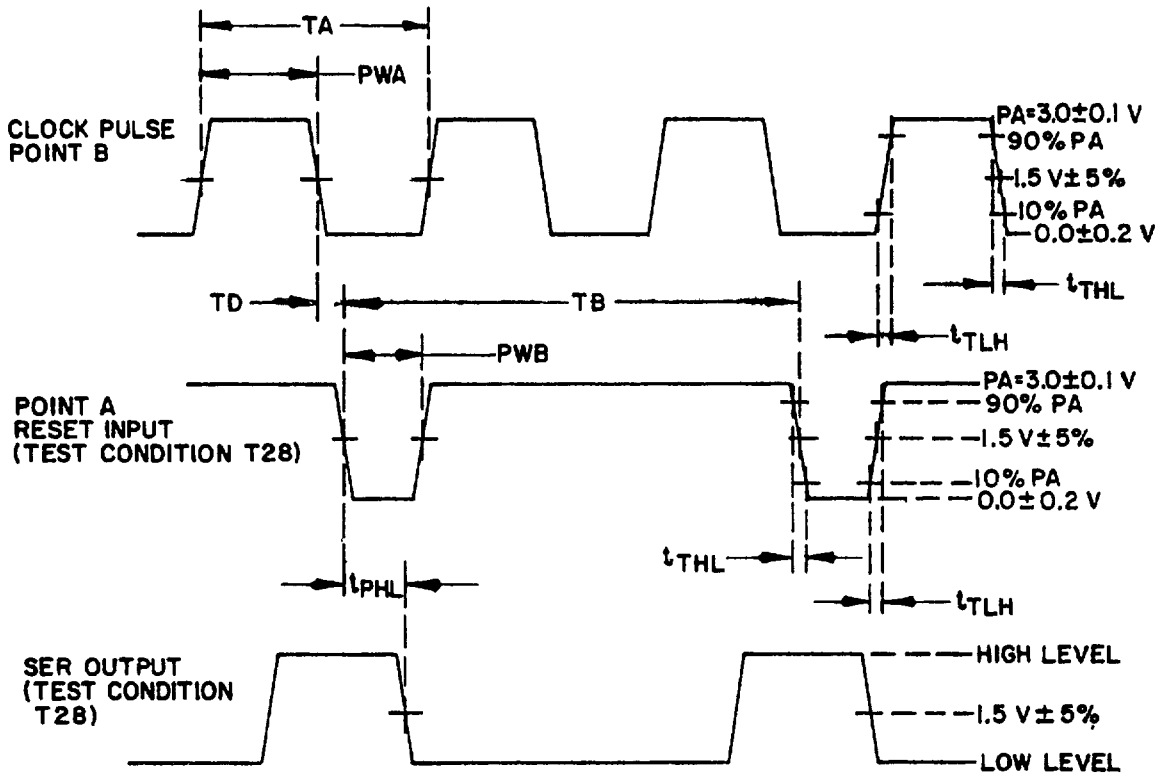
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NOTE. Input and clock pulse parameters for test conditions S25, S26, and T25-T27:
PWA = 10 ±1 µs.
PWB = 5 ±0.5 µs.
TA = 20 ±2 µs.
TB = 10 ±1 µs.
TD = 2.5 ±0.2 µs.
$t_{TLH} = t_{THL} = 10 \pm 2$ ns.

FIGURE 19. Propagation delay time (t_{PLH} and t_{PHL}) test circuit waveforms for table XII.

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NOTE: Input and clock pulse parameters for test condition T28.
PWA = 5 ±0.5 µs.
PWB = 4 ±0.4 µs.
TA = 10 ±1 µs
TB = 20 ±2 µs.
TD = 500 ±50 ns.
$t_{TLH} = t_{THL}$ = 10 ±2 ns.

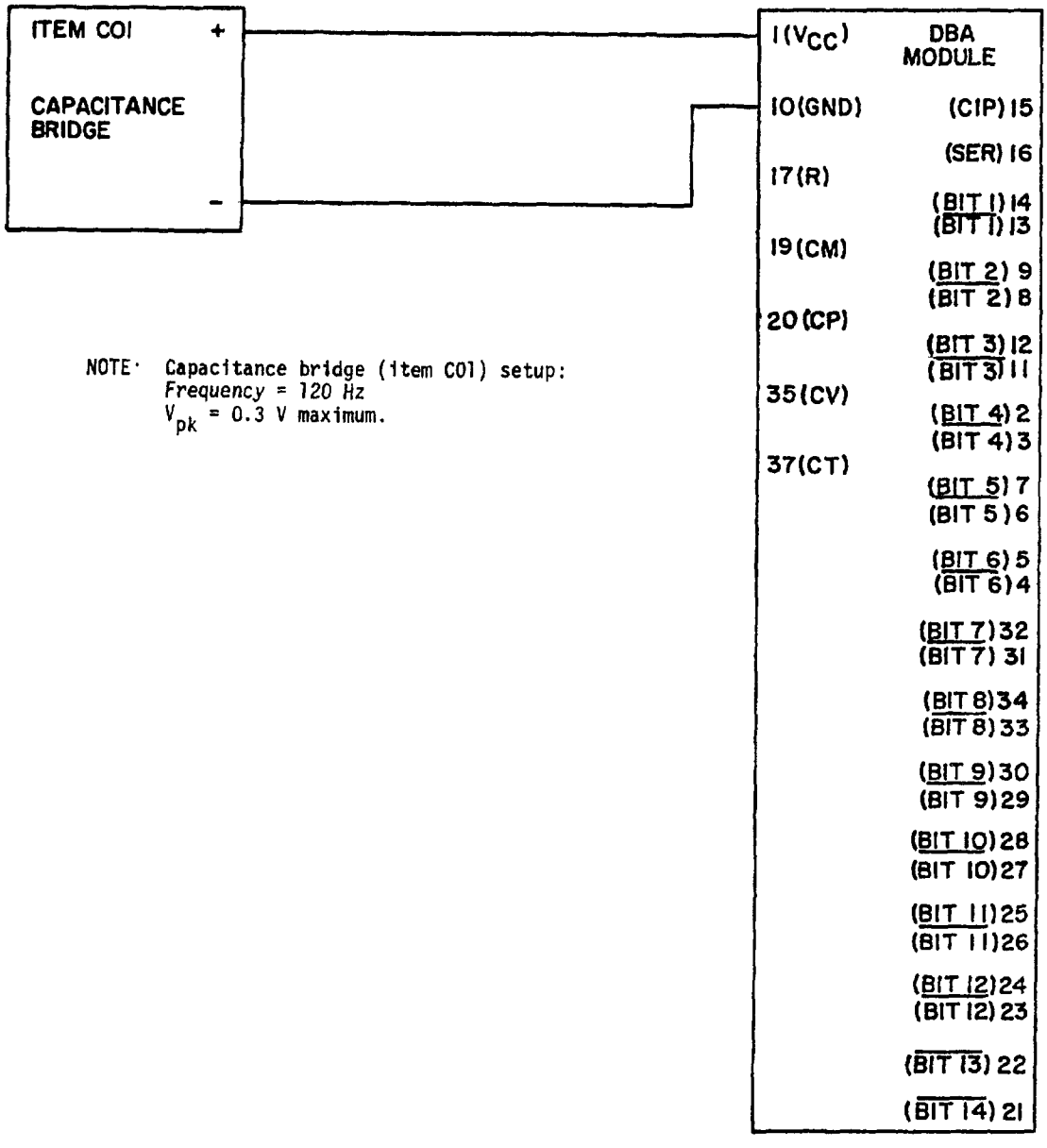
FIGURE 20. Propagation delay time (t_{PHL}) test circuit waveforms for table XII.

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TABLE XIII. Power supply filter capacitance test.

Test condition	Parameter	Initial limits 25° ±5°C Ambient		Unit	Test figure
		Min	Max		
U	Power supply filter capacitance	1.6	3.0	µF	21

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NOTE: Capacitance bridge (item C01) setup:
Frequency = 120 Hz
V_{pk} = 0.3 V maximum.

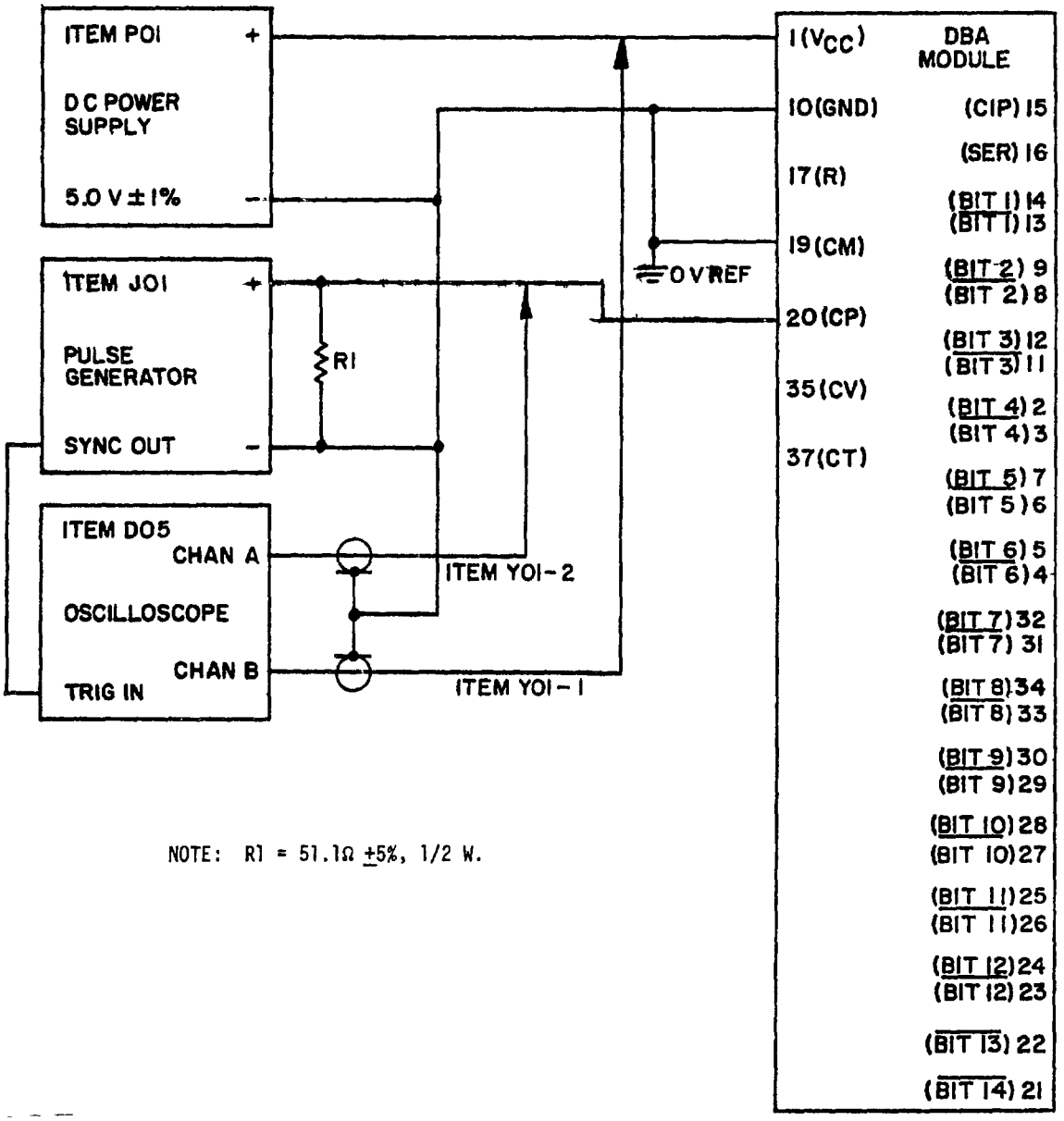
FIGURE 21. Power supply filter capacitance test circuit for table XIII.

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TABLE XIV. Power supply transient amplitude test.

Test condition	Parameter	Connect	Initial limits 25° ±5°C Ambient		Unit	Test figure
		Pin to pin	Min	Max		
V	Power supply transient amplitude	17 to 21	0	400	mV	22, 23

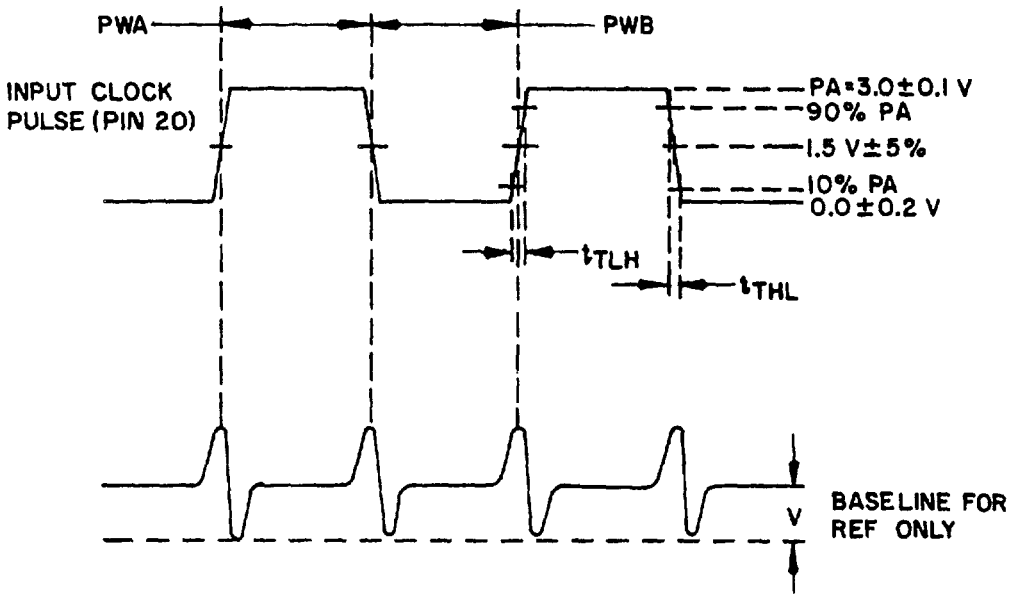
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NOTE: R1 = 51.1Ω ±5%, 1/2 W.

FIGURE 22. Power supply transient amplitude test circuit for table XIV.

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NOTES
1. PWA = 2 ±0.2 µs.
   PWB = 4 ±0.4 µs.
   $t_{TLH} = t_{THL} = 10 \pm 2$ ns.
2. V = Measurement from the baseline to the most negative excursion of the waveform.

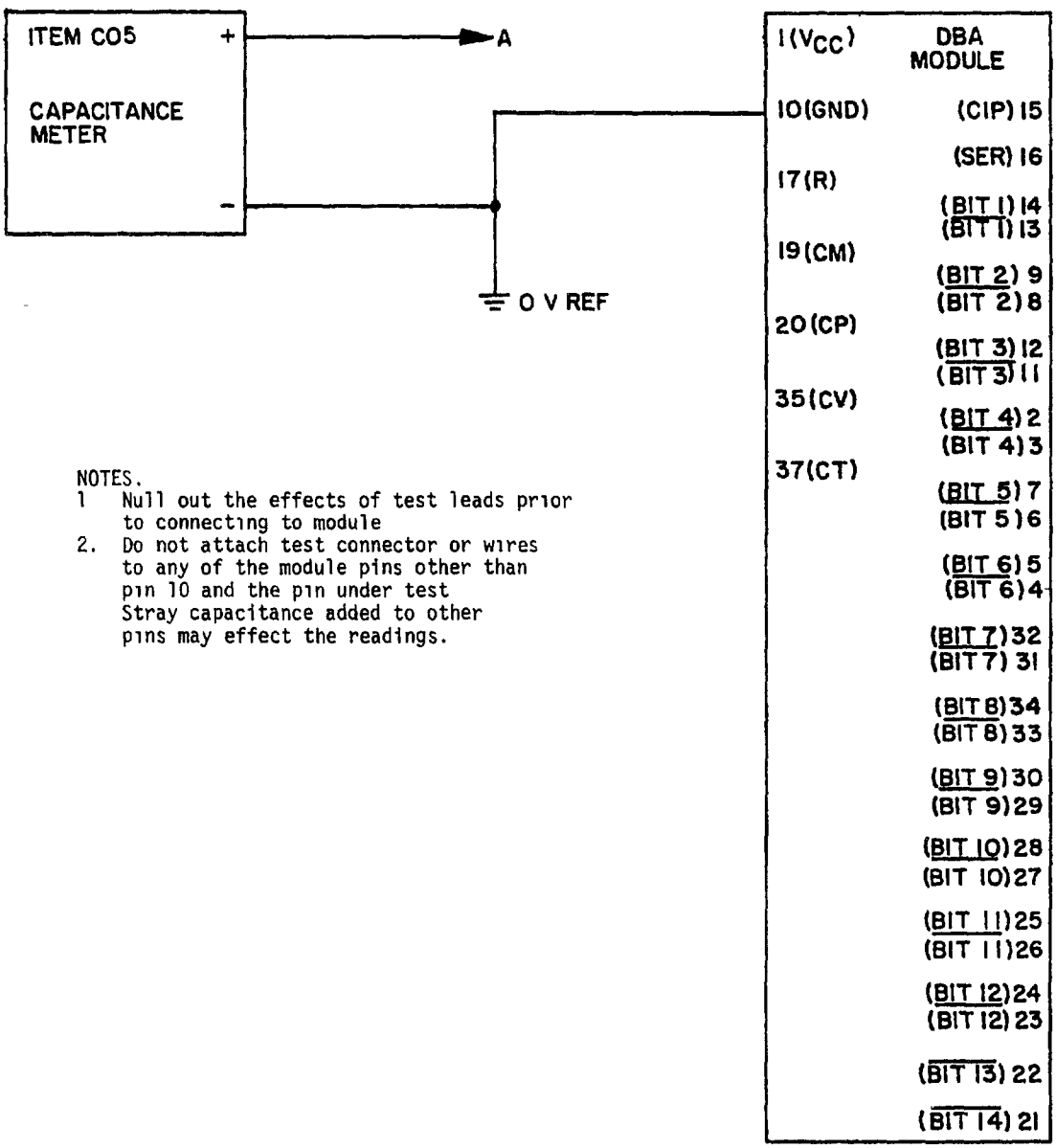
FIGURE 23. Power supply transient amplitude test circuit waveforms for table XIV.

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TABLE XV. Input capacitance test.

Test condition	Parameter		Connect	Initial limits 25° ±5°C Ambient		Unit	Test figure
			A to pin	Min	Max		
W1	CP	Input capacitance	20	2	25	pF	24
W2	CT	↓	37	↓	↓	↓	↓
W3	CV	↓	35	↓	↓	↓	↓
W4	R	↓	17	↓	25	↓	↓
W5	CM	Input capacitance	19	2	15	pF	24

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NOTES.
1 Null out the effects of test leads prior to connecting to module
2. Do not attach test connector or wires to any of the module pins other than pin 10 and the pin under test Stray capacitance added to other pins may effect the readings.

FIGURE 24. Input capacitance test circuit for table XV.

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